

Looked Up or Copied? Later Mentions Decide Which Attention Channel Carries an In-Context Value, and Where an Intervention Appears to Act

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Abstract

A token that states a value in context, such as *shelf* in “the candle is moved to the shelf”, can be read by later tokens through its attention key, matched by a later query, or its value, which is copied. We clamp the two channels separately at this writing token in 10 models from four families and, in a subset of them, on two further tasks. Whenever the candidates are mentioned after the writing token, by an option list or, in larger models, a neutral sentence, the value’s identity is looked up through the key, and in the models we localised the re-mentioned words are the main readers; otherwise the value is mainly copied, even when the same list precedes the writing token. A learned intervention from prior work, whose effect was traced in part to the writing token’s keys in a format that lists the options after the question, has nearly the same behavioural effect in every format. Yet at Qwen2.5-72B its keys alone reproduce 0.89 of its difference from an unfitted control with lettered options, and its values 0.90 with no later mention; across formats, the share its keys carry tracks how much the unpatched model reads the value through keys. In a preregistered test at Mistral-Small-24B, a refit of the intervention with no candidate named after the story shows the same crossover: the format in which it is read, more than the one in which it was fit, decides which component appears to carry it. We separate preregistered from exploratory results and report the predictions that failed.

1. Introduction

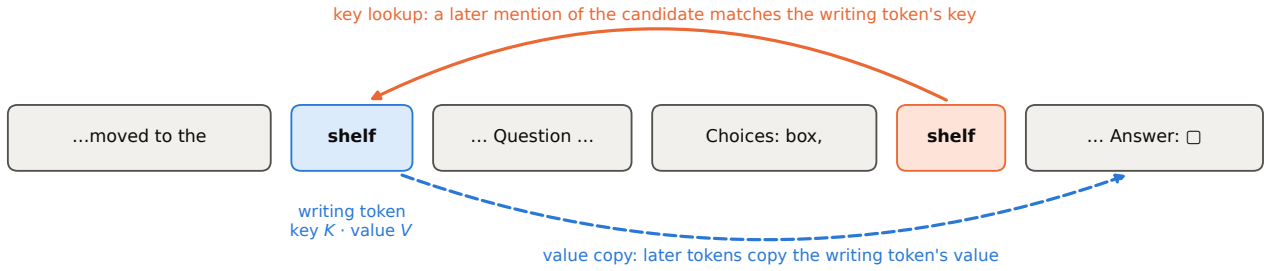
Mechanistic explanations are often stated at the level of components, such as neurons, attention heads or subspaces (Vig et al., 2020; Wang et al., 2022; Geiger et al., 2023); for in-context computation, they say that a value written at one token is read through a particular attention channel, or that an intervention’s effect is carried by a particular component. A later token sees a context token only through its cached key, which a later query must match, and its value, which is copied when the token is attended to. A written value can thus be *looked up* (Figure 1a): a later token whose content matches the key attends to it, so the key itself carries which value was written. Or it can be *copied*: the reader attends for other reasons, and the identity travels in the value. Lookbacks retrieve a state token through bound ordering IDs, in our terms a copy read (Prakash et al., 2026), and binding information can sit in query/key subspaces or, in Llama models, mainly in keys (Oh & Demberg, 2026), but what decides which read a prompt uses is not clear.

In the belief stories of Anonymous (2026), one token, the *writing token* (their “critical token”), states where an object was moved. We clamp its cached key and value separately to those of a run in which a different value was written; clamping both from the first layer reproduces that run exactly (Section 2). We vary whether and where the candidate values are mentioned (Figure 1b), in 10 instruction-tuned models (1.5B–72B, four families) on the belief task, and in five 7–14B models on two further tasks.

Later mentions decide (Section 3). Whenever the candidates are mentioned again after the writing token, by an option list, lettered options or, from 7B up, a neutral sentence, the value’s identity is looked up through the key; without a later mention it is mainly copied through the value. The sentence effect was not predicted; a preregistered, instruction-matched 2×2 design in four 7–14B models then confirmed it and showed that position is causal:

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(a) Two ways later tokens can read the token that wrote a value



(b) Prompt formats (same story; candidates named after, before or not at all)



Figure 1. Two ways later tokens can read a value written in context. (a) In “... moved to the shelf”, the writing token *shelf* (blue) leaves a key K and a value V in the cache at every layer. A later mention of the same candidate, such as *shelf* in an option list after the question (orange), can find it through its key, so the value’s identity is *looked up*. Without such a mention, the identity reaches the answer through the writing token’s value, which is *copied*. We clamp K and V separately to those of another run. (b) The prompt formats share the story and differ in whether and where the candidate values are named, and in the answer instruction: OPTIONS-AFTER, LETTERS-AFTER, SENTENCE-AFTER, NO-MENTION and LIST-BEFORE; the instruction-matched 2×2 of Section 3.1 adds LIST-AFTER and SENTENCE-BEFORE.

the same list or sentence placed before the writing token opens no key read (Table 1). In the 7–14B models we localised, the re-mentioned words are the main readers: under OPTIONS-AFTER the option words alone recover 0.92–0.98 of the key effect at 7–14B, and under SENTENCE-AFTER the sentence’s candidate words recover most of it (Figure 4). The read is an identity lookup in the style of duplicate-token attention (Wang et al., 2022), not a read of belief role (Section 3.4).

Where an intervention appears to act (Section 4). Anonymous (2026) fit a rank-16 distributed alignment search (DAS) intervention (Geiger et al., 2023) that remaps locations. In a format that lists the options after the question, exchanging only the writing token’s keys between it and an unfitted PCA control transferred much of their behavioural difference, which they presented as a component-level explanation of the tested effects. Re-running their released interventions at 24B and 72B, we find that the behavioural effect is nearly the same in every format, while the channel that appears to carry it flips (Figure 2): keys with lettered options, values with no later mention. Our reading, formed after these data, is that the edit is present in both channels and the format’s readers decide which one appears to carry it. Two preregistered tests then supported it: values carry the edit where keys do not, and at Mistral-Small-24B a remap refit with no candidate named after the story shows the same crossover, shifted only where both channels read (Section 4.1). A component-level explanation of this intervention is thus a property of the intervention and the prompt together.

Contributions. (i) Exact, separate key and value clamps at the writing token, an identity measure, and a row-restricted splice that lets only chosen later tokens see a swapped key (Section 2). (ii) Evidence that later mentions decide whether a value is looked up or copied, that their position is causal, and that the re-mentioned words read the value’s identity, not its belief role (Section 3). (iii) A crossover: a learned intervention appears to be carried by keys or by values depending on the format in which it is read, more than the one in which it was fit (refit at Mistral-Small-24B; Section 4). (iv) Six preregistrations, with every main claim classified as confirmatory, exploratory or post hoc (Appendix A); a preregistered transfer law failed at Mistral-Small-24B and held only marginally at Qwen2.5-72B, and a key-share threshold was met in one of three large models (Section 4.2).

2. Setup

Task and prompt formats. We generate belief stories with a port of the released generator of Anonymous (2026): “Everyone initially sees that the $\{o\}$ is in the $\{\text{init}\}$. [...] $\{A\}$ watches as the $\{o\}$ is moved to the $\{\text{LOC}\}$. $\{B\}$ does not see this happen.”, followed by “Where does $\{A\}$ believe the $\{o\}$ is?” (template in Appendix B). Locations are six single-token words. A base story B and a source story S differ only at the *writing token* $\{\text{LOC}\}$ at position p . Two further tasks write the value with one sentence: *paint* (“Later, $\{\text{agent}\}$ repainted the $\{\text{thing}\}$ $\{\text{colour}\}$.”, six colours) and *schedule* (“Later, $\{\text{agent}\}$ moved the $\{\text{thing}\}$

to {day}.”, seven days). Every prompt is wrapped as a system turn, a user turn, the generation prompt with thinking disabled and an “Answer:” prefill, and we read full-vocabulary log-probabilities at the answer position. The formats differ in where, if anywhere, the candidates are named and in the answer instruction (Appendix B): OPTIONS-AFTER (options after the question, “Answer with exactly one choice”, the format of Anonymous 2026); LETTERS-AFTER (options A–F after the question, “Answer with one letter”, scored on the letters); and, all with “Answer with one word”, SENTENCE-AFTER (the neutral sentence “The room has a box, a basket, ... and a closet.” after the story), NO-MENTION (no candidates) and LIST-BEFORE (the options list before the story). On the new tasks the neutral sentence is task-specific and the schedule letters run to G. A *later mention* is any token after p that names a candidate. The instruction-matched 2×2 of Section 3.1 adds LIST-AFTER (OPTIONS-AFTER with the one-word instruction) and SENTENCE-BEFORE (the sentence before the story).

Clamping the key and value channels. A later token sees the writing token only through its cached key, which decides how much the token attends to p , and its value, which decides what is copied when it does. We call these the *key channel* and the *value channel*. The clamp $\mathcal{C}_K(Z)$ runs the base prompt but sets, at p and in every layer $\ell \geq \ell_0$, the pre-RoPE k_{proj} output (all KV groups) to the one cached from run Z ; $\mathcal{C}_V(Z)$ does the same for v_{proj} , and $\mathcal{C}_{KV}(Z)$ does both. Unless stated otherwise $\ell_0 = 0$, and then $\mathcal{C}_{KV}(S)$ reproduces the source run at every later position.¹

Measures. With $m_{\text{SB}} = \log p(S) - \log p(B)$ at the answer position, the effect of clamping channel C is measured against the *self-clamp row* $\mathcal{C}_{KV}(B)$, computed in the same batch:

$$d_C = m_{\text{SB}}(\mathcal{C}_C(S)) - m_{\text{SB}}(\mathcal{C}_{KV}(B)), \quad (1)$$

for $C \in \{K, V, KV\}$. The key share is $s_K = \bar{d}_K / (\bar{d}_K + \bar{d}_V)$, a ratio of means over stories. A foreign key can move m_{SB} without carrying the source value, for instance by weakening B. To isolate identity we also clamp from a story with a third value X that is absent from the story, and with $\Delta_Y(Z) = \log p(Y | \mathcal{C}_K(Z)) - \log p(Y | \mathcal{C}_{KV}(B))$ we define

$$\text{ID}_K = \frac{1}{2} [\Delta_S(S) - \Delta_S(X) + \Delta_X(X) - \Delta_X(S)]. \quad (2)$$

¹Positions before p have identical inputs in B and S and hence identical keys and values; at p both are set to S’s in every layer; by induction over layers every later position receives the same inputs as in S. The clamp precedes key normalisation (Qwen3, OLMo-2) and RoPE, which both runs apply identically at p . We confirm this numerically (Appendix C).

Any effect shared by all foreign keys cancels, so ID_K (in nats) is zero if keys carry no identity of the value; ID_V is defined in the same way with values. Because nats grow with the model’s confidence, we also report the scale-free *identity key share* $s_{\text{ID}} = \overline{\text{ID}}_K / (\overline{\text{ID}}_K + \overline{\text{ID}}_V)$, which, unlike s_K , excludes non-specific effects. The *row-restricted splice* runs attention twice per layer and splices output rows by a mask, so that only a chosen group of later tokens sees the swapped key; it is exact (in an FP32 unit test, all rows reproduce the full key clamp and no rows the clean run, to within 10^{-4} in the logits; Appendix C). The *role-swap control* swaps the observer and non-observer names in the event while keeping the location word, which moves the belief answer from LOC to the initial location; f_C is the fraction of this change carried by the writing token’s channel C alone.

The intervention of Anonymous (2026). Anonymous (2026) patch Mistral-Small-24B and Qwen2.5-72B at the output of (1-based) decoder block 4 over every token of the event span (from “{A} watches as” to the end of the story). For a rank-16 basis U with orthonormal rows, each span position becomes

$$h \leftarrow h_B + U^\top U (h_S - h_B). \quad (3)$$

In the learned remap M , U is a distributed alignment search basis (Geiger et al., 2023) fit with a six-choice cross-entropy objective so that the patched model ranks $T = \pi(S)$ first, where π is a fixed pair-swap of the six locations; it was fit without the “Answer:” prefill of our formats, which Anonymous (2026) add only at evaluation. In the control P, U is an unfitted PCA basis of source-minus-base differences, which transfers S. We reuse the released bases (three seeds each; the Qwen2.5-72B PCA bases are reconstructions, labelled so by Anonymous (2026)). Following their design, a *key-only exchange* replaces the writing token’s key in every block from (1-based) block 6 on by the key of the other patched run, while queries, values and residual states evolve freely; we add the symmetric *value-only exchange*. Unlike our clamp, these exchanges are not exact, because the patch also changes the other span tokens. With $m_{\text{TS}} = \log p(T) - \log p(S)$, writing $P + K_M$ for P with M’s keys and $m(\cdot)$ for m_{TS} in a run,

$$\varphi = \frac{m(M) - m(P)}{m(T) - m(S)}, \quad (4)$$

$$\psi_K = \frac{m(P + K_M) - m(P)}{m(M) - m(P)}, \quad (5)$$

$$\psi_V = \frac{m(P + V_M) - m(P)}{m(M) - m(P)}. \quad (6)$$

φ is the shift of M beyond P towards T, as a fraction of the natural $S \rightarrow T$ shift; ψ_K and ψ_V are the fractions of

the learned-minus-PCA difference that M 's keys or values add to P . The removal fractions $\rho_K = [m(M+K_P) - m(M)]/[m(P) - m(M)]$ and ρ_V (the same with V_P) give M the keys or values of P instead (Appendix C). All are ratios of means over the 96 of their 120 native story cores in which B , S and T are distinct, after averaging the three fits within each core. As a preregistered gate, the argmax token of M and of P under OPTIONS-AFTER must agree with their released outputs on at least 0.95 of the (core, fit) items of each model, over all 120 cores; agreement is 0.994–1.000.

Models, statistics and preregistration. We study 10 instruction-tuned models from four families: Qwen2.5-Instruct (Yang et al., 2024) at 1.5B, 3B, 7B, 14B, 32B and 72B, Qwen3-8B, Mistral-7B-Instruct-v0.3, Mistral-Small-24B-Instruct-2501 (Mistral AI, 2025) and OLMo-2-1124-7B-Instruct (Team OLMo et al., 2024), in BF16 on GPU. We use $n = 150$ stories per format and model. The row-restricted splice uses $n = 60$ at 7–14B; at Qwen2.5-1.5B it ran, like preregistration A, on CPU in FP32 with $n = 40$. Intervals are 95% percentile intervals from 10,000 bootstrap resamples of stories, with ratios recomputed within each resample. Six preregistrations (A–F) were each committed to version control before their run, with the scoring script committed before its outputs were inspected. Appendix A gives every prediction with its recorded outcome, including the failed ones, close to the wording of the log, with format names mapped to ours, and classifies every main claim as confirmatory, exploratory or post hoc.

3. How later tokens read a written value

Preregistered results name their prediction (B1 is prediction 1 of preregistration B; Appendix A); the others are marked exploratory or post hoc.

3.1. Later mentions decide which channel carries the value

Natural formats. Table 8 gives ID_K for every format in all 10 models, and Figure 5 summarises it relative to OPTIONS-AFTER (both in the appendix). The formats fall into three regimes.

Candidates listed after the writing token. Under OPTIONS-AFTER and LETTERS-AFTER, ID_K is large and positive in every model, from 2.8 nats at Qwen2.5-1.5B to 39.0 nats at Qwen2.5-14B under OPTIONS-AFTER. This was preregistered at 1.5B on fresh stories (A1) and in five held-out models from 3B to 8B (B1); both were met. At Qwen2.5-14B and at 24–72B it is exploratory. Under LETTERS-AFTER, the value channel then carries little identity: ID_V is at most about one nat in every model (exploratory; Table 10).

No later mention. Under NO-MENTION, ID_K is at most

Table 1. Position decides the key read. Instruction-matched 2×2 : the same list or neutral sentence naming every candidate, placed after or before the story, with “Answer with one word” in every format; NO-MENTION for reference. Entries are ID_K in nats with 95% bootstrap CIs over $n = 150$ stories, at Qwen2.5-7B and 14B, Mistral-7B and OLMo-2-7B; keys are clamped from layer 0. Preregistration E (prediction E1).

Format	Qwen-7B	Qwen-14B	Mistral-7B	OLMo-2-7B
LIST-AFTER	+20.6 [20.2, 21.0]	+35.8 [34.9, 36.7]	+15.6 [15.1, 16.0]	+8.7 [8.4, 8.9]
SENTENCE-AFTER	+5.5 [5.2, 5.8]	+12.1 [11.5, 12.7]	+7.9 [7.3, 8.4]	+2.0 [1.8, 2.2]
LIST-BEFORE	-0.1 [-0.2, -0.1]	-0.1 [-0.2, -0.0]	+0.0 [-0.1, 0.1]	-0.3 [-0.4, -0.2]
SENTENCE-BEFORE	-0.0 [-0.1, 0.1]	-0.1 [-0.2, 0.0]	-0.0 [-0.1, 0.1]	-0.2 [-0.3, -0.2]
NO-MENTION	+1.0 [0.9, 1.2]	+1.1 [0.9, 1.2]	+0.6 [0.5, 0.6]	+0.4 [0.3, 0.5]

1.25 nats, less than a tenth of its OPTIONS-AFTER value in every model, and the paired OPTIONS-AFTER–NO-MENTION difference is positive in 5/5 held-out models (B2, met). The identity is copied instead: ID_V under NO-MENTION exceeds ID_K more than tenfold in every model (exploratory; Table 10). The residual key identity is small but not zero; in four models from 7B up (Qwen2.5-7B, 14B and 32B and Qwen3-8B) its interval extends just beyond the ± 1 -nat equivalence margin that preregistration A set at 1.5B, which we report as an unpredicted observation. LIST-BEFORE names the same candidates as OPTIONS-AFTER, but before the writing token, where the causal mask prevents them from attending to it. Its ID_K is never reliably positive: it is negative in nine of the ten models, most strongly at Qwen2.5-72B (−2.0 nats), and indistinguishable from zero at Mistral-7B. The ≤ 0.5 -nat bound was preregistered for the three largest models (C4, second part) and met. We have no tested explanation for the negative sign.

A neutral sentence after the story. SENTENCE-AFTER names all six candidates in a plain sentence and asks for a free-form answer. At Qwen2.5-1.5B and 3B it carries no positive key identity: it is slightly negative at 1.5B (−0.5 nats) and indistinguishable from zero at 3B (Table 8). From 7B up it carries an intermediate amount in every model, from 2.0 nats at OLMo-2-7B to 12.1 at Qwen2.5-14B. This was not predicted, and it refutes the narrower “listed options only” account we had formed post hoc at 1.5B (Section 6, limitation ii). Across all formats, with clamps from layer 0, the effects of clamping the key and the value add up, with a small interaction (exploratory; Table 10). The scale-free identity key share s_{ID} compares models fairly: above 1.5B it is 0.94–1.00 under LETTERS-AFTER, 0.67–0.88 under OPTIONS-AFTER, at most 0.45 under SENTENCE-AFTER, at most 0.06 under NO-MENTION and at most 0.00 under LIST-BEFORE (exploratory; Table 9).

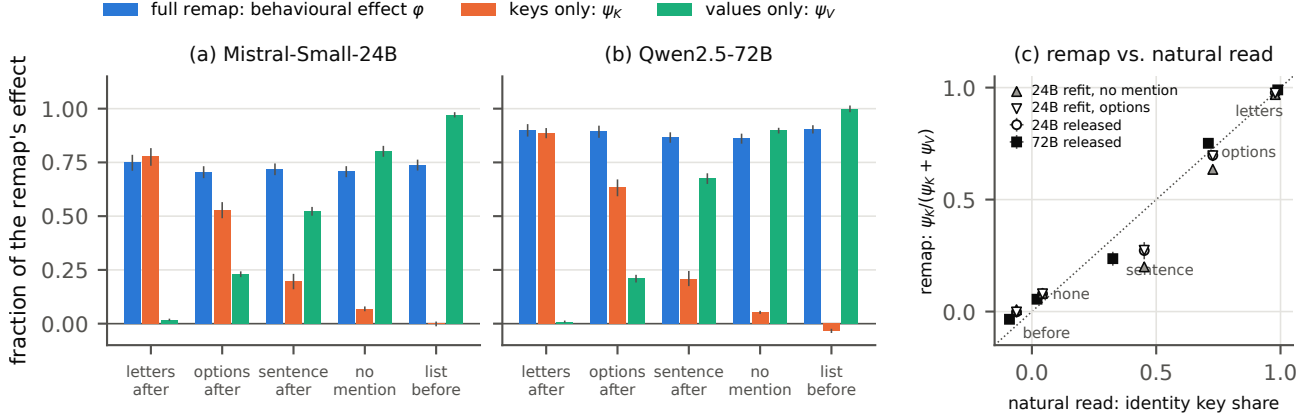


Figure 2. The same learned intervention has nearly the same behavioural effect in every format, but whether the writing token’s keys or its values carry it depends on the later mentions. The rank-16 DAS remap M of Anonymous (2026) against their unfitted PCA patch P (released bases) at Mistral-Small-24B (a) and Qwen2.5-72B (b). Blue: φ , the shift of M relative to P towards the target T , as a fraction of the natural source-to-target shift (Equation (4)). Orange: ψ_K , the fraction of the $M-P$ difference that P reproduces when given M ’s keys at the writing token (Equation (5)). Green: ψ_V , the same with M ’s values (Equation (6)). Keys or values are exchanged from block 6 on. (a, b) $n = 96$ story cores with B, S and T distinct, $\times 3$ fits averaged within core; error bars are 95% bootstrap intervals over cores. (c) The remap’s key share $\psi_K/(\psi_K + \psi_V)$ against the identity key share s_{ID} of the unpatched model in the same format (our stories, same pinned models; dotted line: equality): post hoc for the released remaps (circles, squares), preregistered (F4) for the two refits at Mistral-Small-24B (triangles; Section 4.1). All values, with ρ_K and ρ_V , are in Table 16 (released) and Table 7 (refits).

An instruction-matched 2×2 . The natural formats differ in more than position: OPTIONS-AFTER asks for “exactly one choice”, and a list is not a sentence. Preregistration E therefore crosses {list, sentence} with {before, after}, with the same one-word instruction in every format (Table 1). All parts of E1 were met. ID_K is 8.7–35.8 nats under LIST-AFTER, close to the OPTIONS-AFTER value in the same model, and 2.0–12.1 under SENTENCE-AFTER. Under LIST-BEFORE and SENTENCE-BEFORE, every CI contains zero or lies below it. The paired differences LIST-AFTER–LIST-BEFORE and SENTENCE-AFTER–SENTENCE-BEFORE are positive, with CIs excluding zero, in 4/4 models. Moving the same words from before the writing token to after it switches the key read on, for a list and for a sentence alike. Position, not the instruction or the list format, decides whether the key channel opens. The list format changes only how far it opens: in all four models ID_K is larger under LIST-AFTER than under SENTENCE-AFTER, a comparison we did not preregister.

3.2. Across tasks, families and scale

Tasks. On the paint and schedule tasks, in five models of 7–14B parameters (Table 13), ID_K under OPTIONS-AFTER is 6.5–42.7 nats (paint) and 4.5–28.5 nats (schedule). The paired OPTIONS-AFTER–NO-MENTION difference is positive in 5/5 models per task (D1, met). Under LIST-BEFORE, ID_K is at most +0.32 and +0.16 nats, within the preregistered 0.5-nat bound in 10/10 task–model cells (D2, met). SENTENCE-AFTER again lies between NO-MENTION and

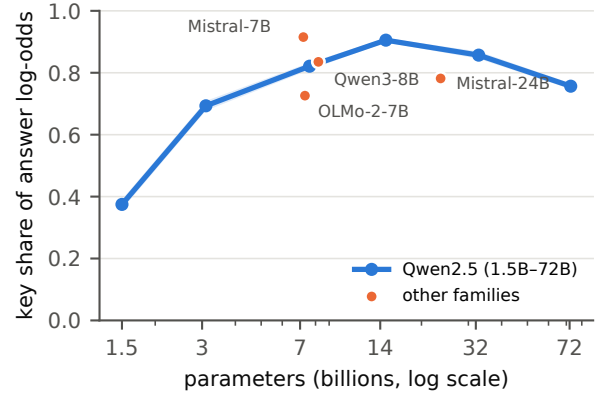


Figure 3. Key share of the OPTIONS-AFTER answer across scale and families. $s_K = \bar{d}_K/(\bar{d}_K + \bar{d}_V)$ with keys and values clamped from layer 0. Line and band: Qwen2.5-Instruct 1.5B–72B with 95% bootstrap CIs; points: other families. $n = 150$ stories per model; values with CIs in Table 8. s_K , the preregistered measure, includes non-specific effects; the identity key share s_{ID} shows the same rise to 14B and fall beyond (Table 9).

OPTIONS-AFTER in every model (exploratory).

Scale and families. Within Qwen2.5, the OPTIONS-AFTER key share rises from 0.37 at 1.5B through 0.69 (3B) and 0.82 (7B) to 0.91 at 14B, then falls to 0.86 at 32B and 0.76 at 72B (Figure 3); the 14B and 72B intervals do not overlap. The size trend was exploratory (B3), and the preregistered threshold $s_K \geq 0.80$ for the three largest mod-

els was met only at 32B (C4; Section 4.2). Re-running Qwen2.5-14B and 32B each in the other’s software and GPU setup gave 0.91 and 0.86, the originals to two decimals (E3, met). The fall beyond 14B is therefore not an artefact of the environment, and it is also seen in the identity key share (0.88 at 14B, 0.71 at 72B), although we do not know its cause. The other families give similar shares (0.73–0.92; Figure 3), and every model except Qwen2.5-1.5B carries more than half of the OPTIONS-AFTER effect in the key.

Depth. Layer-0 keys are a function of the input embedding alone. An exploratory onset sweep (Table 11) starts both clamps deeper: at $\ell_0 \approx 0.06L$ the shares are unchanged in eight of ten models; at $\ell_0 \approx 0.3L$ the share stays at or above one half in seven (up to 0.83) but falls to 0.19, 0.37 and 0.03 at Qwen2.5-1.5B, Mistral-24B and OLMo-2-7B. There the clamp is no longer exact, so these shares are descriptive, and deeper keys still see the written word.

3.3. Which later tokens read the writing token’s key

Which later tokens read the writing token’s key? With the row-restricted splice (Section 2), we let only one group of later tokens see the source key (Figure 4). Under OPTIONS-AFTER, the option words alone recover 0.92–0.98 of the full key effect at 7–14B. The question, the story after the writing token and the remaining tokens, including the answer position, each recover at most 0.01. Under SENTENCE-AFTER, the candidate words of the neutral sentence recover 0.67–0.73 and the other later tokens 0.05–0.11; 0.19–0.22 is not recovered by any single group, so the groups do not add up (Table 15). Both parts of D3 were met in 3/3 models. We did not localise the read at 24B or 72B. The answer position thus gains almost nothing from the source key itself; that it then reads the later mentions we did not test.

An exploratory run at Qwen2.5-1.5B agrees, and places the read in middle layers (Appendix E).

3.4. Identity, not belief role

A later *shelf* attending to an earlier *shelf* is the pattern of duplicate-token heads (Wang et al., 2022). The key read is an identity lookup of this kind and we claim nothing more: “identity” means the lexical identity of a single word in its context, the read is not confined to the lowest layers’ keys (Section 3.2), and we did not identify the heads.

Could the writing token’s channels carry more than identity, such as whether the person asked about saw the move? The role-swap control exchanges the observer’s and the non-observer’s names but keeps the location word. This shifts the belief answer by 12.5 to 79.8 nats, depending on model

and format (Table 14). Clamping the writing token’s key or value alone to the role-swapped run carries at most 0.02 of that shift, under OPTIONS-AFTER, NO-MENTION and LETTERS-AFTER, in five models including Qwen2.5-72B. The prediction was preregistered for OPTIONS-AFTER and NO-MENTION in four models, with a bound of 0.15 (E2, met); LETTERS-AFTER and Qwen2.5-72B are exploratory. The writing token’s key and value thus carry *which* value was written, not the observer’s name; an ordering-ID address of the observer, the same in both runs, would not register (Appendix F).

4. Where an intervention appears to act

Anonymous (2026) read their key exchange between M and P as causal evidence that keys induced by the learned patch contribute to the behavioural difference between the two, under their tested multilayer intervention and the prompt format we call OPTIONS-AFTER. Our runs of their released bases reproduce their saved answers under OPTIONS-AFTER (preregistered gate; Section 2). We then change only the format and add a value-only exchange. Two comparisons are confirmatory and met in both models: C3, $\psi_K(\text{NO-MENTION}) < 0.5 \psi_K(\text{OPTIONS-AFTER})$, and E4, $\psi_V(\text{NO-MENTION}) \geq 0.5$ and $> \psi_K(\text{NO-MENTION})$; the rest of this paragraph and the next is exploratory (Table 16).

Behaviour barely changes across formats; the carrying channel flips. φ lies between 0.70 and 0.75 at Mistral-Small-24B and between 0.86 and 0.91 at Qwen2.5-72B, and the PCA control transfers the source location in every format. Yet at Qwen2.5-72B, M’s keys reproduce 0.89 of the M–P difference under LETTERS-AFTER and 0.63 under OPTIONS-AFTER, but only 0.21 under SENTENCE-AFTER, 0.05 under NO-MENTION and -0.03 under LIST-BEFORE, while M’s values do the reverse ($\psi_V = 0.01, 0.21, 0.67, 0.90, 1.00$). Mistral-Small-24B and the removal fractions ρ_K and ρ_V show the same crossover. Across the ten format–model cells, the remap’s key share $\psi_K/(\psi_K + \psi_V)$ tracks the identity key share s_{ID} of the unpatched model in the same format, measured on our stories at the same pinned revisions (Figure 2c; $r = 0.98$; post hoc for these remaps, preregistered for the refits below).

Interpretation. In LETTERS-AFTER, OPTIONS-AFTER, SENTENCE-AFTER and NO-MENTION the prompt is identical up to the end of the story, so under the causal mask M writes the same key and the same value at the writing token in all four; only their readers differ. The edit is therefore present in both channels, and which one appears to carry it depends on which channel the format’s readers use. This reading was formed after the data of preregistration C; its value half (E4) and the refit test below (F) were then preregistered and confirmed. We localised the readers only

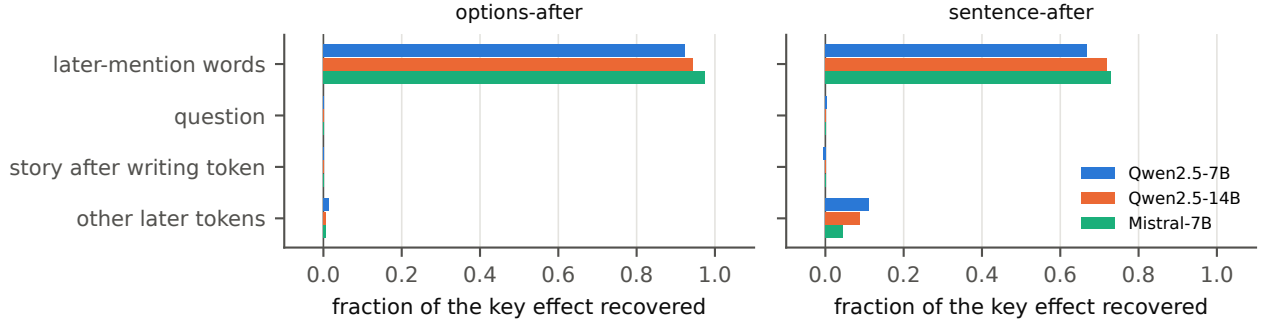


Figure 4. The later mentions read the writing token’s key. Exact row-restricted splice at Qwen2.5-7B, Qwen2.5-14B and Mistral-7B ($n = 60$ stories). Only the listed group of later tokens sees the source key at the writing token, in every layer, while all other tokens see the base key. Bars give the fraction of the full key effect (all tokens see the source key) that this recovers, as a ratio of mean log-odds changes. Left, OPTIONS-AFTER: the later-mention words are the listed options. Right, SENTENCE-AFTER: they are the candidate words of the neutral sentence. “Other later tokens” covers the instruction, chat template, prefill and answer position. Preregistration D (prediction D3); values in Table 15.

at 1.5–14B (Section 3.3), so that they are the same at 24B and 72B is an inference. The addition fractions sum to less than one and the removal fractions mostly to more than one (Table 16), so the channels do not combine additively, and part of the edit may travel through the other patched span tokens. In their format we reproduce their key result: M’s keys alone reproduce a substantial part of the M–P difference in both models ($\psi_K = 0.53$ and 0.63) and, in a comparison they did not make, more of it than M’s values do. They scoped their claim to their tested setting, which must include the prompt. We did not re-test their colour-lookup questions, and score log-odds where they scored argmax agreement.

4.1. A remap fit without later mentions

The remap edits the residual stream at block 4, upstream of every exchanged key and value. Do key readers decode an edit fit without them? If the readers decide, a remap fit where no later mention reads the value through the key should still show the crossover; if the fitting format decides what it writes, it should stay value-carried with lettered options. Preregistration F tested this at Mistral-Small-24B with the released training code of Anonymous (2026), run unmodified through a wrapper that replaces its prompt (for NO-MENTION only), integrity check and model loader (Appendix D): once under NO-MENTION (fit_none) and once, as a same-code control, under their own format (fit_opt). Three seeds each vary only the order of the training pairs; all fits share one PCA initialiser and control P. All gates and all four predictions were met (Table 5; values in Table 7). fit_none is carried by keys under LETTERS-AFTER ($\psi_K = 0.77$, $\psi_V = 0.02$) and by values under NO-MENTION ($\psi_V = 0.80$); it transfers to lettered options ($\varphi = 0.67$); its crossover matches the control’s (difference in differences $-0.008 [-0.027, +0.009]$); and its key share

tracks the natural read ($r = 0.96$; Figure 2c). Post hoc, the fitting format still shifts the two mixed formats in the direction the alternative predicts: fit_none’s ψ_K is lower than the control’s by $0.09 [0.07, 0.12]$ under OPTIONS-AFTER and $0.07 [0.05, 0.10]$ under SENTENCE-AFTER, and it transfers better under NO-MENTION ($\varphi = 0.81$ against 0.72) and worse under LETTERS-AFTER (0.67 against 0.77). The format in which a remap is read decides which channel dominates; the format in which it was fit shifts the balance where both channels read.

4.2. Preregistered predictions that failed

A preregistered transfer law predicted that without a later mention the remap transfers only as far as the value channel carries it: $\varphi(\text{NO-MENTION})$ within ± 0.15 of a value-channel completeness κ_V estimated beforehand by clamping (C1), and $\varphi(\text{OPTIONS-AFTER}) > \varphi(\text{NO-MENTION})$ (C2). C1 fails at Mistral-Small-24B ($\varphi = 0.71 [0.68, 0.73]$ against $\kappa_V = 0.53$) and holds only at the edge of its margin at Qwen2.5-72B; C2 is null at Mistral-Small-24B and small at Qwen2.5-72B ($+0.034 [+0.018, +0.049]$). A key-share threshold $s_K \geq 0.80$ under OPTIONS-AFTER (C4) held only at Qwen2.5-32B of the three large models (Figure 3), while its LIST-BEFORE control held in all three (Appendix A).

5. Related Work

Multiple-choice mechanisms. Attention heads select among listed options: “correct letter” heads with enumeration indices in query/key subspaces (Lieberum et al., 2023), middle-layer attention behind the answer symbol (Wiegraffe et al., 2024), select-and-copy heads (Tulchinskii et al., 2024), decision heads that copy the attended option (Sun et al., 2026), and a winner chosen before it is bound to a

symbol (Wong et al., 2026). Options placed before the context cannot attend to it (Ok & Lee, 2026), which our LIST-BEFORE control exploits. These works ask how an answer is chosen among options; we ask how later mentions read a value written earlier.

Binding, lookbacks and in-context retrieval. Models bind entities to attributes with binding ID vectors (Feng & Steinhardt, 2023). In belief stories, ordering IDs bound at the state token (our writing token) let lookbacks retrieve the queried state’s ordering ID and then the state token itself; these variables are localised in the residual stream, not in the cached key and value (Prakash et al., 2026). Rebinding signals sit in query/key subspaces or mainly in keys, by model family (Oh & Demberg, 2026); an operation’s cache is read mainly as an address rather than a payload (Wu & Shomali, 2026); a router selects which frame is read from a shared value slot (Steele et al., 2026a;b); and editing a field’s cached keys and values barely changes a decision later tokens already recorded (Li, 2026). We ask which of the writing token’s two channels carries the value’s identity, and find that later mentions decide.

Validity of interventions. Interchange interventions align causal variables with representations (Geiger et al., 2021; 2023), but may act through dormant pathways (Makelov et al., 2023; Wu et al., 2024) or planted ones (Liu & Chen, 2026), need not reproduce natural activations (Shih et al., 2026), and are not identifiable from behaviour (Venkatesh & Kurapath, 2026). Diagnostics include control tasks (Hewitt & Liang, 2019), benchmarks (Huang et al., 2024; Arora et al., 2024; Mueller et al., 2025) and input partitions (Li et al., 2026).

Format dependence. Performance is sensitive to meaning-preserving format changes (Sclar et al., 2023); function vectors from open-ended and multiple-choice formats are nearly orthogonal (Opiefka et al., 2026); and a steering gain under one answer encoding does not identify what the intervention controls (Gao et al., 2026). Refusal steering acts mainly through the OV circuit (Cheng et al., 2026), and introspection depends on injected key and value changes (Zou et al., 2026).

Patching and KV-cache edits. Component attribution rests on causal mediation and patching (Vig et al., 2020; Wang et al., 2022), whose conclusions depend on metric and corruption (Zhang & Nanda, 2023; Heimersheim & Nanda, 2024). Precomputed KV caches exactly substitute for a joint pass, and value edits can steer where key arithmetic breaks coherence (Pustovit, 2026); the same causal-mask argument makes our joint clamp exact.

6. Discussion

A practical rule. Which channel carries a written value is set by the tokens that follow the writing token, not by that token alone. A component-level claim about in-context computation, about a circuit or an intervention, should therefore name the prompt structure under which it was measured, above all whether and where the candidate values are mentioned after the writing token, and should be tested under a format with a later mention and one without. Moving the same candidates before the writing token (LIST-BEFORE, SENTENCE-BEFORE) is a cheap control: it keeps the words, the question and the answer position but removes the readers (Table 1). Appendix F relates our results to address-versus-payload accounts of lookback (Prakash et al., 2026; Wu & Shomali, 2026).

Implications for evaluation and attribution. A mechanism localised with the candidates listed after the written value, as in multiple-choice benchmarks (Mueller et al., 2025), may route it through a different channel than the free-form behaviour it should explain. We infer, but have not tested, that attribution methods that freeze attention patterns would miss the key channel, which QK attributions (Kamath et al., 2025) may capture.

Limitations. (i) The tasks are synthetic and templated, and every value is a single token. (ii) Our account was refined post hoc: a “listed options only” account formed and confirmed at 1.5B fails from 7B up, and the later-mention account that replaced it was then tested by preregistration (D3, E1). (iii) A single-channel clamp pairs a source key with a base value, a state no input produces; the joint clamp is exact and ID_K cancels non-specific effects, but such states may be off-distribution. (iv) The fall in key share beyond 14B within Qwen2.5 is unexplained; we have ruled out only the software environment. (v) The small residual key identity under NO-MENTION and the negative ID_K under LIST-BEFORE on the belief task (Section 3.1) are unexplained. (vi) The revisit of Anonymous (2026) covers one kind of intervention at two models and one question, and the refit, which changed only the fitting prompt, one model.

Conclusion. Across four model families and three tasks, later mentions of the candidates decide whether a written value is looked up through its key or copied from its value; where we localised the read, the re-mentioned words are the readers. A released intervention has nearly the same behavioural effect in every format we tested, yet the channel that appears to carry it moves from keys to values as later mentions are weakened or removed, and at Mistral-Small-24B a refit without later mentions does the same.

Impact Statement

This work aims to make explanations of language-model internals more reliable, by showing that the component that appears to carry a value or an intervention can depend on the prompt under which it is measured. More reliable component-level explanations could help researchers audit and debug models, and could reduce the risk that a mechanism found under one evaluation format is trusted to explain behaviour under another. Our results also caution against over-reading interpretability findings: an explanation validated under one prompt format should not, on its own, be used to certify a model’s behaviour under others. Methods for locating and editing what a model reads from its context could in principle also be used to steer models in unwanted ways, although our experiments use synthetic tasks, and the learned interventions we study are one that had already been released and refits of it made with its authors’ released training code on the same synthetic task.

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A. Preregistration log and claim status

We made six preregistrations, A–F. Each was committed to version control before its run, and each scoring script was committed before its outputs were inspected. Tables 2 to 5 give every prediction with its outcome as recorded in the preregistration log of the repository (`docs/PREREGISTRATION.md`). The anonymised code release adds `PREREGISTRATION_HISTORY.txt`, which lists, without author information, every commit that changed the log or a scoring script, with its time and diff, and every commit that changed the results, experiment code or GPU scripts, so that the order of preregistration, run and scoring can be checked; each result file also records the commit of the code that produced it. We keep the wording of the log, with two changes: format names are replaced by ours, and long entries are shortened without changing their content. In the log, the “critical token” or “state token” is the writing token, identity(K) is ID_K , and l_0 is ℓ_0 . The log rounds independently of the tables in Appendix E, so a few values and intervals differ in the last digit (for example, the interval of the Qwen2.5-32B key share).

Table 2. Preregistrations A and B. A: Qwen2.5-1.5B-Instruct in FP32 on CPU, seed 1 (new story cores), $n = 40$ per format, all items, keys clamped from layer 0, 95% core-bootstrap CIs. Predictions committed at 13c1f1f, run at 71e8489. B: GPU stage 1 on held-out models (Qwen2.5-3B/7B, Qwen3-8B, Mistral-7B-v0.3, OLMo-2-7B), $n = 150$ per format, all items, 95% core-bootstrap CIs. Predictions committed at 13c1f1f, code at f706ef5, scoring script `analysis/stage1_prereg.py` committed before inspection at 004579c.

	Prediction	Observed	Verdict
A1	identity(K) > 1.0 nats with a CI excluding 0, in OPTIONS-AFTER and in LETTERS-AFTER	OPTIONS-AFTER +2.85 [2.38, 3.33]; LETTERS-AFTER +4.10 [3.45, 4.77]	met
A2	identity(K) CI within the equivalence margin ± 1.0 in NO-MENTION, LIST-BEFORE and SENTENCE-AFTER	NO-MENTION +0.10 [−0.01, 0.21]; LIST-BEFORE −0.20 [−0.28, −0.13]; SENTENCE-AFTER −0.41 [−0.56, −0.26]	met
A3	Key share ($l_0 = 0$) in OPTIONS-AFTER within [0.25, 0.50]	0.38 [0.35, 0.41]	met
B1	identity(K) in OPTIONS-AFTER and LETTERS-AFTER positive, CI excluding 0, in at least 4 of 5 models	5/5 models	met
B2	identity(K)(OPTIONS-AFTER) − identity(K)(NO-MENTION) > 0, CI excluding 0, in at least 4 of 5 models	5/5 models	met
B3	Exploratory, no directional claim: how the OPTIONS-AFTER key share ($l_0 = 0$ and $l_0 = 0.0625 L$) varies with size within Qwen2.5	Rises steadily with size: 1.5B 0.37 [0.36, 0.39]; 3B −0.69 [0.68, 0.71]; 7B 0.82 [0.81, 0.83]; 14B 0.91 [0.90, 0.91]. The depth-matched l_0 gives identical values. Recorded later, with the outcome of C: non-monotonic beyond 14B.	–

Notes recorded with outcome A. LIST-BEFORE and SENTENCE-AFTER are slightly but reliably negative. They sit inside the margin but are not exactly zero. The seed-0 LETTERS-AFTER interaction (+2.0 [0.3, 3.7]) did not replicate: seed 1 gives +0.6 [−1.0, 2.2]. This confirms the post hoc option-listing claim in the same model on new data; generalisation across models is preregistration B.

Unpredicted observations recorded with outcome B (reported, not reinterpreted). SENTENCE-AFTER carries large key identity in the larger models (Qwen2.5-7B +5.5, Qwen2.5-14B +12.1, Qwen3-8B +8.8, Mistral-7B +7.9, OLMo-2-7B +2.0 nats) but about 0 at Qwen2.5-1.5B/3B (−0.47 and −0.04). So the post hoc “listed answer options only” refinement from 1.5B does not generalise. LIST-BEFORE is at or below 0 in every model (−0.45 to +0.01). NO-MENTION carries small positive identity(K) in larger models (+1.0 to +1.25 nats at 7–14B), outside the CPU-era ± 1 margin, though under 5% of OPTIONS-AFTER.

History of the claim, including post hoc steps. The account of when keys carry the value was revised twice after seeing data. We list every step.

1. *Starting hypothesis.* Before the first format factorial, our hypothesis was that any re-mention of the candidates after the writing token opens the key channel.
2. *Post hoc narrowing at 1.5B.* In the first run at Qwen2.5-1.5B (seed 0, $n = 40$), SENTENCE-AFTER showed no key effect. We therefore narrowed the claim to “listed answer options after the writing token open the key channel”. This narrowing was made after seeing the data, and an internal go/no-go gate written before that run was not met as written.
3. *Preregistration A* tested the narrowed claim on fresh story cores in the same model, and all three predictions were met. This confirms the narrowed claim in one model on new data, not across models.

Table 3. Preregistration C: GPU stage 2, the released bases of [Anonymous \(2026\)](#) at Mistral-Small-24B and Qwen2.5-72B across formats, and the natural factorial at 24–72B. Predictions and the scoring script `analysis/stage2_score.py` committed before any stage-2 run (5dc82f5); code at 2bleca7. C4 is scored by `analysis/stage1_prereg.py` (committed at 004579c). Primary population: native story cores with B, S and T distinct; fits averaged within core; 95% core bootstrap. κ_V was estimated before the run from the fixed-value data released with [Anonymous \(2026\)](#). Here ψ is our ψ_K .

Prediction	Mistral-Small-24B	Qwen2.5-72B
Gate: in OPTIONS-AFTER, the argmax of M and of P agrees with the outputs released with Anonymous (2026) on ≥ 0.95 of (core, fit) items	M 0.994, P 1.000 ($n = 360$ each): passed	M 1.000, P 1.000: passed
C1, transfer law: $\varphi_{\text{NO-MENTION}}(\text{M})$ within ± 0.15 of κ_V	0.708 [0.681, 0.734] vs 0.53 ± 0.15 : not met	0.861 [0.837, 0.884] vs 0.73 ± 0.15 : met (at the edge)
C2: $\varphi_{\text{OPTIONS-AFTER}}(\text{M}) - \varphi_{\text{NO-MENTION}}(\text{M}) > 0$, paired CI excluding 0	−0.003 [−0.019, +0.013]: not met	+0.034 [+0.018, +0.049]: met (small)
C3: $\psi_{\text{NO-MENTION}} < 0.5 \cdot \psi_{\text{OPTIONS-AFTER}}$	0.069 vs 0.528: met	0.052 vs 0.633: met
Prediction (natural factorial)	Outcome	
C4: OPTIONS-AFTER key share ≥ 0.80 at Mistral-Small-24B, Qwen2.5-32B and Qwen2.5-72B	not met overall: Mistral-24B 0.78 [0.77, 0.79] no; Qwen2.5-32B 0.86 [0.84, 0.87] yes; Qwen2.5-72B 0.76 [0.75, 0.77] no.	
C4: LIST-BEFORE identity(K) ≤ 0.5 nats in all three	met in all three: −0.53, −0.42, −1.99.	

4. *Preregistration B* tested held-out models. Both directional predictions were met in 5/5 models, but from 7B up SENTENCE-AFTER carries an intermediate amount of key identity, so the “listed options only” account does not generalise. The account we now give, that from 7B up a later mention of the candidates opens the key channel, whether a list or a neutral sentence (a list most strongly), while at 1.5B and 3B only a list does, was revised again, post hoc, after stage 1. It is closer to the starting hypothesis. The residual NO-MENTION identity in the Qwen models from 7B to 32B (up to 1.25 nats) extends just beyond the ± 1 -nat margin that A set at 1.5B; in the other models it stays inside. We report it as unpredicted, not as a confirmation.
5. *Preregistration C* added a transfer law derived from the fixed-value data of [Anonymous \(2026\)](#). It failed at Mistral-Small-24B (C1 and C2 not met) and held only marginally at Qwen2.5-72B. C3 was met in both models. The key-share threshold of C4 was met only at Qwen2.5-32B; its LIST-BEFORE control was met in all three models. The reading that the intervention is written into both channels, and that the format decides which one appears to carry it, was formed after seeing these data.
6. *Preregistration D* retested the options-list part of the account on two new tasks (D1, D2) and localised the readers at 7–14B (D3). It contains no test of whether SENTENCE-AFTER opens the key channel; SENTENCE-AFTER on the new tasks was exploratory.
7. *Preregistration E* was written in response to two internal reviews of a draft. It tested the revised account with an instruction-matched 2×2 (E1), including the first confirmatory test that a sentence after the story opens the key channel (E1c, SENTENCE-AFTER − SENTENCE-BEFORE). It also tested the value half of the post hoc reading of C (E4), the role control (E2) and the software environment (E3). All its predictions were met.
8. *Preregistration F* tested the other half of that reading: whether the format in which a remap is read, rather than the one in which it was fit, decides which channel appears to carry it, by refitting the remap without later mentions at Mistral-Small-24B. All gates and predictions were met; a smaller shift of the key share in the two mixed formats, in the direction the alternative predicts, was found post hoc.

Table 6 lists each main claim with its status. “Confirmatory” means that a preregistered prediction tested the claim in the stated models and was met. “Exploratory” means that the measurement was planned or run without a prediction. “Post hoc” means that the claim was formed after seeing the data it rests on.

Table 4. Preregistrations D and E. D: GPU stage 3; predictions and the scoring script `analysis/stage3_score.py` committed before any stage-3 run (0f5cbff); run at b8815cb. New tasks: $n = 150$ per format in Qwen2.5-7B, Qwen2.5-14B, Qwen3-8B, Mistral-7B-v0.3 and OLMo-2-7B; localisation: belief task, $n = 60$, in Qwen2.5-7B, Qwen2.5-14B and Mistral-7B, the order of the three D3 values. E: GPU stage 3b; predictions and the scoring script `analysis/stage3b_score.py` committed before any stage-3b run, which ran at that commit (3ca63f1). E1 and E2 use Qwen2.5-7B, Qwen2.5-14B, Mistral-7B and OLMo-2-7B; E4 uses Mistral-Small-24B and Qwen2.5-72B.

	Prediction	Result	Verdict
D1	Paint: per model, $\text{OPTIONS-AFTER ID}_K > 0$ and $\text{OPTIONS-AFTER} - \text{NO-MENTION} > 0$ (CIs excluding 0); met if ≥ 4 of 5 models count	5/5 models ($\text{OPTIONS-AFTER ID}_K +6.5$ to $+42.7$ nats)	met
D1	Schedule: same	5/5 models ($\text{OPTIONS-AFTER ID}_K +4.5$ to $+28.5$ nats)	met
D2	LIST-BEFORE $\text{ID}_K \leq 0.5$ nats in ≥ 9 of the 10 task $\times$ model cells	10/10 task $\times$ model cells (max $+0.32$)	met
D3	OPTIONS-AFTER: choice-word rows recover ≥ 0.70 of the full key effect; question rows and the remaining tail rows each ≤ 0.15 ; in all 3 models	3/3 models: choice words 0.92 / 0.94 / 0.98; question 0.00; tail ≤ 0.01	met
D3	SENTENCE-AFTER, where the full key effect is positive: re-mention-word rows recover ≥ 0.50 in ≥ 2 of 3 models	3/3 models: 0.67 / 0.72 / 0.73	met
E1a	Instruction-matched 2×2 , every arm “Answer with one word”: $\text{ID}_K(\text{LIST-AFTER}) > 0$, CI excluding 0, in 4/4 models	$+8.7$ to $+35.8$ nats, 4/4	met
E1b	$\text{ID}_K(\text{LIST-BEFORE}) \leq 0.5$ and $\text{ID}_K(\text{SENTENCE-BEFORE}) \leq 0.5$, each in $\geq 3/4$ models	4/4 and 4/4 (max $+0.01$)	met
E1c	Paired LIST-AFTER $-$ LIST-BEFORE > 0 in 4/4 models; paired SENTENCE-AFTER $-$ SENTENCE-BEFORE > 0 in $\geq 3/4$; each CI excluding 0	4/4; 4/4	met
E2	Role control: for the direct question in OPTIONS-AFTER and NO-MENTION, $ f_K \leq 0.15$ and $ f_V \leq 0.15$ in $\geq 3/4$ models (Qwen2.5-72B exploratory)	4/4 and 4/4 (all fractions ≤ 0.02 ; also ≤ 0.01 at Qwen2.5-72B)	met
E3	Environment: each re-run within 0.03 of the original OPTIONS-AFTER key share (Qwen2.5-14B under transformers 5.9.0 on 2 GPUs, original 0.91; Qwen2.5-32B under transformers 5.18.0 on 1 GPU, original 0.86)	Qwen2.5-14B: 0.91 (original 0.91); Qwen2.5-32B: 0.86 (original 0.86)	met
E4	Value-only exchange in the frames of Anonymous (2026): $\psi_V(\text{NO-MENTION}) \geq 0.5$ and $\psi_V(\text{NO-MENTION}) > \psi_K(\text{NO-MENTION})$ in both models	Mistral-24B 0.80 vs 0.07; Qwen-72B 0.90 vs 0.05	met

Exploratory parts of D and E, with no prediction: key shares on the new tasks; LETTERS-AFTER and SENTENCE-AFTER identity on the new tasks; layer-window and KV-group localisation at 7–14B; ψ_V and ρ_V in the other formats; the role control in LETTERS-AFTER and at Qwen2.5-72B. Before preregistration E, an exploratory CPU pilot of the role control was run at Qwen2.5-1.5B ($n = 40$; recorded in the log with E).

Table 5. Preregistration F. GPU stage 4: the remap of [Anonymous \(2026\)](#) refit at Mistral-Small-24B with their released training code, under NO-MENTION (fit_none) and under their own format (fit_opt, same-code control). Predictions, gates and the scoring script `analysis/stage4_score.py` committed before the run, which ran at that commit (`1ef8696`). Thresholds refer to point estimates except F3; every statistic is per run, with the three seeds averaged within core; 96 cores, 95% core bootstrap.

	Prediction or gate	Result	Verdict
G1	Each refit P spans the released P of the same seed: mean squared principal cosine ≥ 0.99	1.000 for all six (one P for all seeds)	met
G2	$\varphi \geq 0.5$ under each run’s fitting prompt	fit_none 0.81; fit_opt 0.72	met
G3	fit_opt reproduces the released crossover: $\psi_K(\text{LETTERS-AFTER}) \geq 0.5$, $\psi_V(\text{NO-MENTION}) \geq 0.5$	0.78, 0.81	met
F1	fit_none: $\psi_K(\text{LETTERS-AFTER}) \geq 0.5$ and $\psi_V(\text{NO-MENTION}) \geq 0.5$	0.77, 0.80	met
F2	fit_none: $\varphi(\text{LETTERS-AFTER}) \geq 0.5$	0.67	met
F3	$D = [\psi_K(\text{LETTERS-AFTER}) - \psi_K(\text{NO-MENTION})]_{\text{fit_none}} - [\cdot]_{\text{fit_opt}}$, paired over cores: CI within $[-0.25, +0.25]$	$-0.008 [-0.027, +0.009]$	met
F4	For each run, $\psi_K/(\psi_K + \psi_V)$ over the five formats correlates with the natural s_{ID} of Mistral-Small-24B at $r \geq 0.9$	fit_none 0.96; fit_opt 0.98	met

Stated alternative. If the fitting format decided which channel carries the remap, F1 and F2 would fail with $\psi_V(\text{LETTERS-AFTER}) \geq 0.5$ for fit_none and $D \leq -0.4$. This all-or-none pattern, which needs an edit that values can read and keys cannot, was not observed; the graded post hoc shift in OPTIONS-AFTER and SENTENCE-AFTER (Section 4.1) is the more sensitive comparison. *Exploratory parts of F*, with no prediction: evaluation without the “Answer:” prefix; ρ_K and ρ_V ; principal cosines between the remaps; loss curves; the identity-objective fits that the training code makes alongside the remap.

Looked Up or Copied? Later Mentions Decide Which Attention Channel Carries an In-Context Value

Table 6. Status of each main claim. Section of the main text, claim, and whether it is confirmatory (with the prediction that tests it), exploratory or post hoc.

Section	Claim	Status
Section 3.1	Candidates listed after the writing token (OPTIONS-AFTER, LETTERS-AFTER) give a large positive ID_K	Confirmatory at 1.5B (A1), in five held-out 3–8B models (B1) and, for LIST-AFTER, in four 7–14B models (E1a); exploratory at Qwen2.5-14B under OPTIONS-AFTER and LETTERS-AFTER and at 24–72B
Section 3.1	Without a later mention (NO-MENTION), keys carry little identity	Confirmatory as a paired contrast with OPTIONS-AFTER (B2) and at 1.5B (A2); the residual in the Qwen models from 7B to 32B (up to 1.25 nats), just beyond A’s ± 1 -nat margin, is unpredicted
Section 3.1	Without a later mention, the identity is copied through the value channel ($ID_V \gg ID_K$)	Exploratory (Table 10)
Section 3.1	Candidates before the writing token (LIST-BEFORE, SENTENCE-BEFORE) carry no positive key identity ($ID_K \leq 0.5$ nats in every model and task tested; ≤ 0 in most models on the belief task, down to -2.0 nats at Qwen2.5-72B)	Confirmatory: A2 at 1.5B (within ± 1 nat), and the ≤ 0.5 -nat bound in C4 second part (24–72B), D2 (new tasks) and E1b (7–14B); the negative sign was not predicted and is unexplained
Section 3.1	SENTENCE-AFTER carries intermediate key identity from 7B up, and no positive key identity at 1.5B and 3B	Post hoc (observed in B); the 1.5B null was predicted by A2 (equivalence margin, met), and the positive part from 7B up was then confirmed in four 7–14B models by E1c (SENTENCE-AFTER – SENTENCE-BEFORE > 0)
Section 3.1	With the instruction matched, moving a list or a sentence from before to after the writing token switches the key read on	Confirmatory in four 7–14B models (E1a–c); that a list reads more than a sentence, and that LIST-AFTER is close to OPTIONS-AFTER, are exploratory
Section 3.1	Key and value effects add up, with a small interaction	Exploratory (Table 10)
Section 3.2	The pattern holds on the paint and schedule tasks	Confirmatory for OPTIONS-AFTER, the OPTIONS-AFTER – NO-MENTION contrast and LIST-BEFORE (D1, D2); exploratory for SENTENCE-AFTER, LETTERS-AFTER and the key share
Section 3.2	Within Qwen2.5, the OPTIONS-AFTER key share rises with size to 14B, then falls	Exploratory (B3); the threshold $s_K \geq 0.80$ failed (C4)
Section 3.2	The fall beyond 14B is not a software-environment artefact	Confirmatory (E3)
Section 3.2	The key read is not only the input embedding	Exploratory (onset sweep, Table 11)
Section 3.3	The later-mention words read the key at 7–14B	Confirmatory for OPTIONS-AFTER and SENTENCE-AFTER in three 7–14B models (D3); exploratory for LETTERS-AFTER
Section 3.3	At 1.5B, option words read the key in layers 4–15, one KV group dominant	Exploratory (no prediction)
Section 3.3	The answer then reads the later mentions	Hypothesis; not tested
Section 3.4	The writing token’s channels carry identity, not belief role	Confirmatory for OPTIONS-AFTER and NO-MENTION in four models (E2); exploratory for LETTERS-AFTER and Qwen2.5-72B
Section 4	Reproduction of Anonymous (2026) under OPTIONS-AFTER	Confirmatory gate (C), passed
Section 4	ψ_K is much smaller without a later mention	Confirmatory (C3)
Section 4	Without a later mention, values carry the intervention	Confirmatory (E4)
Section 4	φ varies little across formats (0.70–0.75 at Mistral-Small-24B, 0.86–0.91 at Qwen2.5-72B); the full crossover of ψ_K and ψ_V over five formats	Exploratory for the released remaps (the refit test is F); the one preregistered contrast (C2, OPTIONS-AFTER – NO-MENTION) was null at Mistral-Small-24B and small but positive at Qwen2.5-72B (+0.034 [+0.018, +0.049])
Section 4	The intervention is written into both channels; the format decides which appears to carry it	Post hoc reading of C; then confirmed for its value half (E4) and for a remap fit without later mentions (F1–F4)
Section 4	The released remaps’ key share tracks the natural identity key share ($r = 0.98$)	Post hoc, exploratory; preregistered only for the refits (F4)
Section 4.1	A remap fit under NO-MENTION shows the same crossover as one fit under OPTIONS-AFTER: the format in which it is read decides which channel dominates	Confirmatory at Mistral-Small-24B (F1–F4)
Section 4.1	The fitting format shifts the key share in OPTIONS-AFTER and SENTENCE-AFTER, and transfer under NO-MENTION and LETTERS-AFTER	Post hoc (key share) and exploratory (φ)
Section 4.2	The transfer law from clamping-estimated κ_V fails	Confirmatory test, failed at Mistral-Small-24B (C1, C2) and marginal at Qwen2.5-72B

B. Prompts

Belief task. The story generator (`ckeys/story.py`) is ported from the released generator of [Anonymous \(2026\)](#). Each story core draws an agent and an other agent (Alice, Bob), an object and a distractor object from twenty objects, and four locations from the six candidates *box, basket, shelf, drawer, cabinet, closet*. The initial location, the base location and the source location must differ, and the distractor’s location must differ from the base and source locations. The two “initially sees” sentences are sorted by object name. The direct question is “Where does $\{A\}$ believe the $\{o\}$ is?”. Base B and source S differ only at the writing token, the word after “moved to the”. The third value X used for ID_K and ID_V is drawn, with a seed fixed by the core, from the candidates other than the base, source, initial and distractor locations, so it appears nowhere in the story. We keep a core only if the B, S and X prompts have the same token length and differ at exactly one token.

We show the first core generated with seed 0: agent Alice, other agent Bob, object *lamp*, distractor *coin*, initial location *basket* (the distractor is also in the basket), base location *cabinet*, source location *drawer*, third value *box*. Below are the raw user-turn prompts of the base run in all seven formats, as built by `ckeys/encoding.py`. Line breaks inside the story are added here for width; in the prompts the story is one line.

OPTIONS-AFTER (the format of the key-exchange experiments of [Anonymous 2026](#)).

Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone initially sees that the lamp is in the basket. Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Choices: box, basket, shelf, drawer, cabinet, closet
Answer with exactly one choice.
Answer:

LIST-AFTER. As OPTIONS-AFTER, with the last instruction line replaced by “Answer with one word.”.

...
Question: Where does Alice believe the lamp is?
Choices: box, basket, shelf, drawer, cabinet, closet
Answer with one word.
Answer:

LETTERS-AFTER.

Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone initially sees that the lamp is in the basket. Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.
Question: Where does Alice believe the lamp is?
Choices: A) box, B) basket, C) shelf, D) drawer, E) cabinet, F) closet
Answer with one letter.
Answer:

SENTENCE-AFTER.

Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone initially sees that the lamp is in the basket. Alice watches as the lamp is moved to the cabinet. Bob does not see this happen. The room has a box, a basket, a shelf, a drawer, a cabinet and a closet.
Question: Where does Alice believe the lamp is?
Answer with one word.
Answer:

SENTENCE-BEFORE.

Read the story and answer the question.

Story: The room has a box, a basket, a shelf, a drawer, a cabinet and a closet. Everyone initially sees that the coin is in the basket. Everyone initially sees that the lamp is in the basket. Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.

Question: Where does Alice believe the lamp is?

Answer with one word.

Answer:

NO-MENTION.

Read the story and answer the question.

Story: Everyone initially sees that the coin is in the basket. Everyone initially sees that the lamp is in the basket. Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.

Question: Where does Alice believe the lamp is?

Answer with one word.

Answer:

LIST-BEFORE.

Read the story and answer the question.

Choices: box, basket, shelf, drawer, cabinet, closet

Story: Everyone initially sees that the coin is in the basket. Everyone initially sees that the lamp is in the basket. Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.

Question: Where does Alice believe the lamp is?

Answer with one word.

Answer:

Chat wrapper and scoring. Each raw prompt is wrapped as in the `Answer:-prefill` interface of [Anonymous \(2026\)](#) and its released encoder: a system turn “You are a helpful assistant.”, a user turn holding the raw prompt, the model’s generation prompt, and the assistant prefill “Answer:”. We also disable thinking (`enable_thinking=False`, which matters for Qwen3 and is ignored by other templates). The raw prompt itself ends in “Answer:”, so this string appears twice. For Qwen2.5 the full input is

```
<|im_start|>system
You are a helpful assistant.<|im_end|>
<|im_start|>user
Read the story and answer the question.
...
Answer:<|im_end|>
<|im_start|>assistant
Answer:
```

The wrapper asserts that the raw prompt occurs exactly once in the templated text. Templates that reject a system role fall back to merging the system text into the user turn; every output file records whether this happened, and no model reported here needed it. We read full-vocabulary log-probabilities at the last position and score the candidates as single tokens with a leading space (“ box”, “ basket”, ..., or “ A” to “ F” under LETTERS-AFTER); the code asserts that each candidate is one token.

Role-swap control. The role-swapped story R exchanges the two names in the event and keeps the location word. For the core above, the base event and the role-swapped event are

B: Alice watches as the lamp is moved to the cabinet. Bob does not see this happen.

R: Bob watches as the lamp is moved to the cabinet. Alice does not see this happen.

The question still asks for Alice’s belief, so the correct answer moves from the writing token’s location (*cabinet*) to the initial location (*basket*). The role effect is measured with $m = \log p(\text{initial}) - \log p(\text{LOC})$, and f_C is the effect of clamping channel C at the writing token to R’s, divided by the full R–B difference in m .

Intervention frames. The stage-2, stage-3b and stage-4 runs with the bases of [Anonymous \(2026\)](#) use their released native story cores, rendered with the same generator, the same format builders and the same wrapper. The patched span is the move event, from the observer’s name to the end of the story; the writing token is the location word inside it.

Paint and schedule tasks. The two new tasks (`ckeys/tasks.py`) follow the same structure: two sentences give the starting values of two things, sorted by name, and a final sentence writes the new value at the writing token. They use the formats OPTIONS-AFTER, LETTERS-AFTER, SENTENCE-AFTER, NO-MENTION and LIST-BEFORE, built exactly as above with the task’s candidates; the neutral sentence of SENTENCE-AFTER is task-specific. The agent is drawn from six names (Alice, Bob, Carol, David, Emma, Frank) and the things from ten per task.

	paint	schedule
Candidates	red, blue, green, yellow, black, white	Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday
Things	fence, door, bench, shed, gate, chair, table, wall, boat, bike	meeting, dinner, appointment, party, class, interview, concert, match, lecture, trip
Starting value	“The {thing} was painted {value}.”	“The {thing} was scheduled for {value}.”
Writing sentence	“Later, {agent} repainted the {thing} {value}.”	“Later, {agent} moved the {thing} to {value}.”
Question	“What color is the {thing} now?”	“On which day is the {thing} now?”
SENTENCE-AFTER sentence	“The shop sells red, blue, green, yellow, black and white paint.”	“The calendar shows Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday.”
LETTERS-AFTER options	A) red, ..., F) white	A) Monday, ..., G) Sunday

The first cores generated with seed 0 give, under OPTIONS-AFTER and SENTENCE-AFTER,

Read the story and answer the question.

Story: The boat was painted yellow. The shed was painted yellow. Later, Emma repainted the shed black.

Question: What color is the shed now?

Choices: red, blue, green, yellow, black, white

Answer with exactly one choice.

Answer:

Read the story and answer the question.

Story: The interview was scheduled for Friday. The match was scheduled for Monday. Later, Frank moved the interview to Wednesday. The calendar shows Monday, Tuesday, Wednesday, Thursday, Friday, Saturday and Sunday.

Question: On which day is the interview now?

Answer with one word.

Answer:

with source values *green* and *Thursday* and third values *blue* and *Tuesday*.

C. Implementation and checks

Clamp hooks. Each clamp is a forward hook on the `k_proj` and/or `v_proj` module of every decoder block $\ell \geq \ell_0$. The hook overwrites the output row at the writing position p with the vector cached from the donor run (B, S or X) at the same layer. These outputs come before RoPE and before key normalisation (Qwen3, OLMo-2), and they contain all KV groups, so under grouped-query attention every group is clamped. All clamp conditions for one story run as rows of a single batch of the base prompt: the self-clamp row, $\mathcal{C}_K$, $\mathcal{C}_V$ and $\mathcal{C}_{KV}$ to S at each onset ℓ_0 , and the same three to X at $\ell_0 = 0$. The onsets are $\ell_0 = 0$, $\ell_0 = \text{round}(0.0625 L)$, which matches block 6 of 80, where the exchange of Anonymous (2026) starts in Qwen2.5-72B, and $\ell_0 = \text{round}(0.3 L)$. Forward passes use `use_cache=False` and compute logits only at the last position. From preregistration A onward, every natural-factorial output file records the number of skipped cores, which is zero in every such run.

Self-clamp row. The self-clamp row $\mathcal{C}_{KV}(\text{B})$ sets the writing token’s keys and values to their own values, so mathematically it equals the clean base run. In BF16, however, a batched forward pass can differ slightly from an unbatched one. From preregistration A onward we therefore measure every effect against this row of the same batch (Equation (1)), not against a separately run clean base, so that batching noise cancels. Each result summary records the gap between the self-clamp row and the clean run as a noise floor. The first CPU factorials (FP32) and all row-restricted splices measure against the clean base run.

Exactness of the clamp. By the argument in Section 2, $\mathcal{C}_{KV}(\text{S})$ from $\ell_0 = 0$ reproduces the source run at every later position. In the FP32 CPU runs at Qwen2.5-1.5B, the recovery ratio $d_{KV}/[m_{\text{SB}}(\text{S}) - m_{\text{SB}}(\text{B})]$ equals one to the printed precision in every format (`results/format_factorial/`). The unit tests (`tests/test_interventions.py`, Qwen2.5-0.5B in FP32) check that an identity edit and a self key exchange leave the logits unchanged to within 10^{-4} , and that patching the residual stream at the writing token after one block transfers the source answer.

Row-restricted splice. The splice wraps the forward pass of each attention module. It runs the module once with the base key at p and once with the source key, which the `k_proj` hook inserts, and returns the source output in the rows of the chosen token group G and the base output elsewhere. In the layer-window and KV-group variants, the source key is inserted only in the chosen layers, or only in one group’s slice of the key. A unit test checks that $G = \text{all rows}$ reproduces the full key clamp and $G = \emptyset$ reproduces the clean run, each to within 10^{-4} in the logits. Every run also evaluates $G = \emptyset$ for each story and reports its mean effect, which is zero to the printed precision.

A KV-cache pitfall. Our first splice implementation was silently wrong. Transformers 5 builds a `DynamicCache` by default, even for a single forward pass. Running the attention module twice per layer therefore appended keys to the cache, and the second (source-key) pass silently read the base keys. The fix is to run every forward pass with `use_cache=False`. The exactness test above was added in the same change, and every splice result reported here was produced after it. Anyone who patches attention by calling a module more than once per forward pass should check for this.

Exchanges in the intervention frames. The patch of Anonymous (2026) replaces the output of decoder block 4 (1-based) over every token of the move-event span by Equation (3), materialised in BF16 as in their engine, with the released rank-16 bases for the learned remap M and the PCA control P , at seeds 101–103 (and, in stage 4, the refit bases of Appendix D, each with the shared P). The key-only exchange replaces the writing token’s `k_proj` output in blocks 6 to L (1-based) by the other patched run’s, giving $\text{P} + \text{K}_\text{M}$ (addition) and $\text{M} + \text{K}_\text{P}$ (removal); queries, values and residual states evolve freely. The value-only exchange does the same with `v_proj`. The removal fractions are

$$\rho_K = \frac{m(\text{M} + \text{K}_\text{P}) - m(\text{M})}{m(\text{P}) - m(\text{M})}, \quad \rho_V = \frac{m(\text{M} + \text{V}_\text{P}) - m(\text{M})}{m(\text{P}) - m(\text{M})},$$

with $m = m_{\text{TS}}$, averaged over the three fits within each core before the ratio of means is taken. The reproduction gate compares the argmax token of M and P under `OPTIONS-AFTER` with the outputs released with Anonymous (2026).

Code audit. Before the GPU runs, an independent review of the measurement code led to fixes made before any GPU result existed, among them the batched self-clamp baseline, depth-matched onsets, a seeded third value X, core-bootstrap CIs computed in the experiment scripts, the competence filter (now a secondary analysis) and provenance written into every output file.

Mistral-Small-24B tokenizer warning. When loading the Mistral-Small-24B tokenizer, transformers warns that its pre-tokenisation regex is incorrect and suggests `fix_mistral_regex=True`. We kept the default, under which the reproduction gate of preregistration C passed. In stage 4 we checked that the flag leaves the token ids of all 3,600 frames prompts (five formats, both prefills) unchanged; we have not checked the natural-factorial prompts.

Compute. All experiments except the stage-4 refits are forward passes; the refits train only rank-16 bases (Appendix D), and no model weights are ever trained. Pilots with models of at most 1.7B parameters, the first format factorials, preregistration A, the 1.5B row-restricted splices and the attention probe ran on CPU in FP32; the preregistration-A output records transformers 5.18.0 and Python 3.11.15. All GPU runs used NVIDIA A100-SXM4-80GB GPUs in BF16, torch 2.11.0 with CUDA 12.8 and Python 3.12.14, as recorded in each stage’s `ENV.txt` and output files:

- stage 1 (preregistration B; Qwen2.5-1.5B to 14B, Qwen3-8B, Mistral-7B, OLMo-2-7B): one GPU, transformers 5.18.0, commit `f706ef5`;
- stage 2 (preregistration C; frames at 24B and 72B, natural factorial at 24–72B): two GPUs with `device_map="auto"`, transformers 5.9.0, pinned by the stage-2 script for the reproduction gate, commit `2b1eca7`;
- stage 3 (preregistration D; new tasks and localisation at 7–14B): one GPU, transformers 5.18.0, commit `b8815cb`;
- stage 3b (preregistration E): two GPUs, commit `3ca63f1`; the 2×2, the role control at 7–14B and the Qwen2.5-32B environment re-run under transformers 5.18.0 (the re-run on one GPU), and the Qwen2.5-14B environment re-run, the value-only frames and the role control at 72B under transformers 5.9.0;
- stage 4 (preregistration F): two GPUs, one per refit (55 minutes each), then the frames on both, transformers 5.9.0, commit `1ef8696`.

Model revisions. For the two models of [Anonymous \(2026\)](#) we pin the revisions used for the reproduction: Mistral-Small-24B-Instruct-2501 at `9527884be6e5616bdd54de542f9ae13384489724` and Qwen2.5-72B-Instruct at `495f39366efef23836d0cfae4fbe635880d2be31`. The pins apply to the frames and to the stage-2 natural factorial at those sizes. All other runs, including Qwen2.5-32B and the role control at Qwen2.5-72B, load the default revision available at run time.

Released code. The code release contains the story generator, encoder, tasks and hook primitives (`ckeys/`); one script per experiment (`experiments/`); the preregistered scoring scripts (`analysis/stage1_prereg.py`, `stage2_score.py`, `stage3_score.py`, `stage3b_score.py`, `stage4_score.py`); the refit wrapper (`experiments/refit_remap.py`); one shell script per GPU stage, which pins the library version and records the commit and environment (`scripts/`); the exactness tests (`tests/`); and `paper/make_figures.py`, which generates every result figure, every table in Appendices D and E and every result macro in the text from the saved results (the schematic in Figure 1 is drawn by `paper/make_schematic.py`).

D. Refitting the remap (preregistration F)

Recipe. We ran the released training code of [Anonymous \(2026\)](#) (gpu/train.py and gpu/behavior_engine.py, unmodified) through experiments/refit_remap.py. Its recipe: a rank-16 basis at the output of (1-based) block 4 over every event-span token; the pair-swap objective with a six-way conditional cross-entropy at the start of the reply (no prefill); AdamW, learning rate 10^{-3} , batch 1, one shuffled epoch of their 1000 training pairs (2000 stories); seeds 101–103, which change only the order of the training pairs: every fit starts from the same PCA basis of source-minus-base differences (the randomised sketch differs by seed but returns the same subspace, mean squared principal cosine 1.000 across seeds), and that basis is the control P of every fit. The code also fits an identity-objective basis next to each remap (exploratory): its shift of the answer towards T, relative to P and as a fraction of the natural S→T shift, is at most 0.18 for fit_none (under LETTERS-AFTER) and 0.04 for fit_opt. The training stories are disjoint from the 120 evaluation cores, but most evaluation event sentences also occur in some training story, as for the released remap.

What the wrapper replaces. At import time, and recorded with hashes in each run’s FRAME.json: the prompt function, for fit_none only, by our NO-MENTION prompt, which agrees token for token with the original prompt up to the end of the story in all 1000 training pairs; the release-integrity check, by one that tolerates only the release’s mismatched README.md; and the model loader, by one with the same load arguments (BF16, sdpa, eval mode, no gradients, no cache, pad = eos, fix_mistral_regex) at the pinned revision, waiving the release’s exact Python, package, device and memory pins and its model-asset receipt (the weights are identified instead by the pinned commit, which the loader checks). Under the causal mask the activations up to the end of the story are the same in both runs, so both start from the same P; the PCA bases came out bitwise identical across runs.

Results. Table 7 gives every frame quantity for the two refits and for the released bases re-run in the same job; Table 5 gives the gates and predictions. The rest of this paragraph is exploratory. The control matches the released remap to within 0.03 in every quantity and format. Every fitted remap keeps 0.34–0.36 of its subspace (mean squared principal cosine) in the shared PCA start, against 0.003 for random subspaces, so overlaps between fits lie above that floor: fit_opt overlaps the released remap at 0.71–0.79, while fit_none overlaps the released remap at 0.37–0.39 and fit_opt at 0.37–0.39, less than two seeds of one fit overlap each other (0.41–0.44). Without the “Answer:” prefill, which matches the interface the remaps were fit at, the difference in differences is +0.004 [−0.014, +0.021]. The training loss of the remap fits fell from 2.9–4.8 nats over the first 100 updates to 0.11–0.48 over the last 100; it starts above chance ($\ln 6 \approx 1.79$) because the PCA start transfers the source, which the pair-swap objective scores as wrong.

Table 7. **Refit remaps in the frames of Anonymous (2026)** (Mistral-Small-24B, “Answer:” prefill). fit_none: refit under NO-MENTION; fit_opt: refit under OPTIONS-AFTER with the same code; released: the released bases, re-run in the same job. All fits share one PCA control P. $n = 96$ cores, three fits averaged within core; 95% core bootstrap.

Remap	Format	φ	ψ_K	ρ_K	ψ_V	ρ_V
fit_none	LETTERS-AFTER	0.67 [0.64, 0.70]	0.77 [0.73, 0.80]	0.98 [0.97, 0.98]	0.02 [0.02, 0.03]	0.15 [0.13, 0.19]
	OPTIONS-AFTER	0.72 [0.70, 0.74]	0.43 [0.39, 0.47]	0.74 [0.73, 0.76]	0.25 [0.23, 0.26]	0.54 [0.50, 0.58]
	SENTENCE-AFTER	0.81 [0.79, 0.83]	0.12 [0.10, 0.15]	0.47 [0.44, 0.49]	0.50 [0.47, 0.52]	0.85 [0.81, 0.88]
	NO-MENTION	0.81 [0.79, 0.84]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.77, 0.82]	0.93 [0.91, 0.94]
	LIST-BEFORE	0.76 [0.74, 0.78]	0.01 [−0.00, 0.02]	−0.00 [−0.01, 0.01]	0.97 [0.96, 0.99]	0.95 [0.94, 0.96]
fit_opt	LETTERS-AFTER	0.77 [0.74, 0.80]	0.78 [0.73, 0.82]	0.98 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.72 [0.69, 0.74]	0.52 [0.48, 0.56]	0.76 [0.74, 0.77]	0.23 [0.22, 0.24]	0.44 [0.40, 0.48]
	SENTENCE-AFTER	0.73 [0.71, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.72 [0.69, 0.74]	0.07 [0.06, 0.08]	0.17 [0.15, 0.19]	0.81 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.77 [0.74, 0.79]	−0.00 [−0.01, 0.01]	0.00 [−0.01, 0.02]	0.98 [0.96, 0.99]	0.96 [0.95, 0.98]
released	LETTERS-AFTER	0.75 [0.71, 0.79]	0.78 [0.74, 0.82]	0.99 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.70 [0.68, 0.73]	0.53 [0.49, 0.57]	0.76 [0.75, 0.78]	0.23 [0.22, 0.24]	0.44 [0.41, 0.48]
	SENTENCE-AFTER	0.72 [0.69, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.71 [0.68, 0.73]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.74 [0.71, 0.76]	−0.00 [−0.01, 0.01]	0.01 [−0.00, 0.02]	0.97 [0.96, 0.98]	0.96 [0.95, 0.98]

E. Additional results

Unless stated otherwise, intervals are 95% percentile intervals from 10,000 bootstrap resamples of stories (cores), with ratios recomputed within each resample. Each estimate is printed with its interval below it, except in Tables 10 and 11, where the intervals of the individual effects are in the result summaries released with the code.

Natural factorial in all models. Table 8 gives the OPTIONS-AFTER key share and ID_K in every format for all 10 models, and Figure 5 shows ID_K relative to OPTIONS-AFTER. LIST-BEFORE is negative in every model except Mistral-7B, where it is indistinguishable from zero; at Qwen2.5-72B it reaches -2.0 nats.

Table 8. Natural factorial, all models. Key share s_K under OPTIONS-AFTER and value identity carried by keys, ID_K (nats), in each format, with keys and values clamped from layer 0. $n = 150$ stories per format and model; 95% bootstrap CIs over stories (ratio recomputed in each resample for s_K). Models ordered by size. Rows up to 14B are from GPU stage 1, whose predictions B1–B2 cover OPTIONS-AFTER, LETTERS-AFTER and NO-MENTION in Qwen2.5-3B/7B, Qwen3-8B, Mistral-7B and OLMo-2-7B; rows from 24B are from stage 2, whose prediction C4 covers s_K and LIST-BEFORE. The other cells, including the 1.5B and 14B rows, are exploratory; preregistration A tested 1.5B in a separate run (Table 2).

Model	s_K (OPTIONS-AFTER)	ID_K , identity carried by keys (nats)				
		LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
Qwen2.5-1.5B	0.37 [0.36, 0.39]	+3.6 [+3.3, +3.8]	+2.8 [+2.6, +3.1]	-0.5 [-0.6, -0.4]	+0.2 [+0.1, +0.2]	-0.2 [-0.3, -0.2]
Qwen2.5-3B	0.69 [0.68, 0.71]	+11.4 [+10.4, +12.4]	+10.3 [+9.6, +11.1]	-0.0 [-0.4, +0.3]	+0.1 [-0.0, +0.2]	-0.5 [-0.6, -0.3]
Mistral-7B	0.92 [0.90, 0.93]	+15.2 [+14.6, +15.7]	+14.9 [+14.5, +15.3]	+7.9 [+7.3, +8.4]	+0.6 [+0.5, +0.6]	+0.0 [-0.1, +0.1]
OLMo-2-7B	0.73 [0.71, 0.74]	+7.9 [+7.7, +8.0]	+8.2 [+8.0, +8.4]	+2.0 [+1.8, +2.2]	+0.4 [+0.3, +0.5]	-0.3 [-0.4, -0.2]
Qwen2.5-7B	0.82 [0.81, 0.83]	+20.4 [+20.0, +20.8]	+21.2 [+20.7, +21.6]	+5.5 [+5.2, +5.8]	+1.0 [+0.9, +1.2]	-0.1 [-0.2, -0.1]
Qwen3-8B	0.84 [0.83, 0.84]	+33.9 [+33.4, +34.3]	+35.4 [+34.6, +36.2]	+8.8 [+8.3, +9.3]	+1.2 [+1.0, +1.4]	-0.3 [-0.4, -0.3]
Qwen2.5-14B	0.91 [0.90, 0.91]	+34.8 [+34.5, +35.1]	+39.0 [+38.0, +40.0]	+12.1 [+11.5, +12.7]	+1.1 [+0.9, +1.2]	-0.1 [-0.2, -0.0]
Mistral-24B	0.78 [0.77, 0.79]	+9.2 [+9.1, +9.3]	+9.6 [+9.4, +9.8]	+5.6 [+5.4, +5.9]	+0.4 [+0.3, +0.5]	-0.5 [-0.6, -0.5]
Qwen2.5-32B	0.86 [0.85, 0.87]	+34.8 [+34.5, +35.1]	+32.4 [+31.6, +33.2]	+10.3 [+9.6, +10.9]	+1.1 [+0.9, +1.3]	-0.4 [-0.6, -0.3]
Qwen2.5-72B	0.76 [0.75, 0.77]	+26.4 [+26.2, +26.6]	+25.6 [+25.0, +26.2]	+8.4 [+8.0, +8.9]	+0.5 [+0.4, +0.6]	-2.0 [-2.2, -1.8]

Identity key share. Table 9 normalises ID_K by the total identity effect, $s_{ID} = \overline{ID}_K / (\overline{ID}_K + \overline{ID}_V)$, so that formats and models with different confidence can be compared. It tracks s_K closely except under NO-MENTION and LIST-BEFORE, where s_K also counts the non-specific effect of a foreign key.

Table 9. Identity key share s_{ID} , all models (exploratory). Ratio of mean ID_K to mean $ID_K + ID_V$, keys and values clamped from layer 0; $n = 150$ stories per format and model; 95% bootstrap CIs, ratio recomputed in each resample.

Model	LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
Qwen2.5-1.5B	+0.83 [+0.81, +0.86]	+0.36 [+0.34, +0.38]	-0.08 [-0.10, -0.06]	+0.02 [+0.01, +0.02]	-0.04 [-0.05, -0.03]
Qwen2.5-3B	+0.98 [+0.97, +0.99]	+0.67 [+0.66, +0.69]	-0.00 [-0.04, +0.03]	+0.00 [-0.00, +0.01]	-0.05 [-0.06, -0.04]
Mistral-7B	+0.94 [+0.92, +0.95]	+0.87 [+0.85, +0.88]	+0.41 [+0.39, +0.43]	+0.04 [+0.03, +0.04]	+0.00 [-0.00, +0.00]
OLMo-2-7B	+0.99 [+0.99, +0.99]	+0.69 [+0.68, +0.70]	+0.19 [+0.17, +0.20]	+0.03 [+0.03, +0.04]	-0.03 [-0.03, -0.02]
Qwen2.5-7B	+0.99 [+0.98, +0.99]	+0.79 [+0.77, +0.80]	+0.32 [+0.30, +0.33]	+0.06 [+0.05, +0.06]	-0.01 [-0.01, -0.00]
Qwen3-8B	+0.97 [+0.97, +0.98]	+0.81 [+0.80, +0.82]	+0.29 [+0.28, +0.30]	+0.05 [+0.04, +0.05]	-0.01 [-0.02, -0.01]
Qwen2.5-14B	+1.00 [+1.00, +1.00]	+0.88 [+0.87, +0.89]	+0.40 [+0.38, +0.41]	+0.05 [+0.05, +0.06]	-0.00 [-0.01, -0.00]
Mistral-24B	+0.98 [+0.98, +0.98]	+0.73 [+0.72, +0.74]	+0.45 [+0.44, +0.46]	+0.04 [+0.03, +0.05]	-0.06 [-0.07, -0.05]
Qwen2.5-32B	+0.99 [+0.99, +0.99]	+0.82 [+0.81, +0.83]	+0.30 [+0.29, +0.32]	+0.04 [+0.04, +0.05]	-0.01 [-0.02, -0.01]
Qwen2.5-72B	+0.99 [+0.99, +0.99]	+0.71 [+0.70, +0.72]	+0.33 [+0.31, +0.34]	+0.02 [+0.02, +0.02]	-0.09 [-0.10, -0.08]

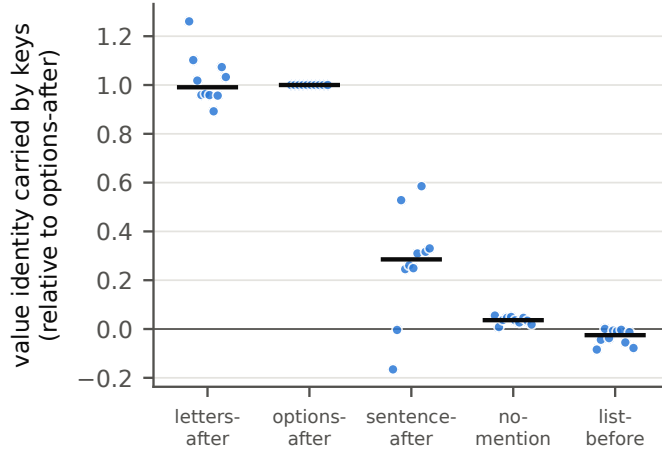


Figure 5. **Key identity by format, relative to OPTIONS-AFTER.** Each point is one of the 10 models: ID_K in that format divided by ID_K under OPTIONS-AFTER in the same model (so OPTIONS-AFTER is 1 by construction); points are jittered horizontally, and the horizontal line is the median over models. $n = 150$ stories per format and model; per-model values with 95% bootstrap CIs are in Table 8.

Decomposition. Table 10 gives, for each model and format, the effects of clamping the key, the value or both, the interaction $d_{KV} - d_K - d_V$, and ID_V . The interaction is small relative to d_{KV} in every cell, so the two channels’ effects add up. Under LETTERS-AFTER the value channel carries almost no identity, and under NO-MENTION and LIST-BEFORE it carries most of it: there the identity is copied.

Clamp onset. Table 11 starts both clamps deeper. With $\ell_0 \approx 0.06L$ the shares are essentially unchanged. With $\ell_0 \approx 0.3L$ the shares stay substantial in most models (up to 0.83 at Qwen2.5-14B), but fall to 0.19 at Qwen2.5-1.5B and to 0.03 at OLMo-2-7B, where the key effect depends on the lowest layers. With $\ell_0 > 0$ the clamp is no longer exact, and at $\ell_0 \approx 0.3L$ the key and value effects no longer add up: the interaction is positive in every model (result summaries). The share at that onset is therefore descriptive only.

Layer windows and KV groups. Table 12 refines the row-restricted splice at Qwen2.5-1.5B (28 layers, two KV groups). Only the option-word rows see the source key, and the swap is further limited to a window of four layers (all KV groups) or to one KV group’s slice of the key (all layers). The read happens in layers 4–15, and KV group 1 carries more of it than KV group 0 in both formats. The seven windows together recover less than the option-word rows do with the swap at all layers, and the two single-group swaps together recover less than both groups at once, so the read compounds across layers and groups. We ran the same window and KV-group splices, on the option-word rows under OPTIONS-AFTER and LETTERS-AFTER, at Qwen2.5-7B, Qwen2.5-14B and Mistral-7B (exploratory; result summaries). In these models no single four-layer window and no single KV group recovers most of the option-word effect, so the dominance of one KV group seen at 1.5B does not carry over.

New tasks. Table 13 gives the paint and schedule results. In every model and both tasks, ID_K is large under LETTERS-AFTER and OPTIONS-AFTER, intermediate under SENTENCE-AFTER, small under NO-MENTION and near zero under LIST-BEFORE. The OPTIONS-AFTER key share ranges from 0.47 to 0.92 (paint) and from 0.35 to 0.92 (schedule), lowest at OLMo-2-7B in both.

Role control. Table 14 gives the role-swap effect and the fractions of it that the writing token’s key and value carry. The role swap shifts the belief answer by many nats in every model and format, and neither channel at the writing token carries more than 0.02 of that shift.

Localisation. Table 15 gives the row-restricted splice for each token group. The later-mention words are the listed options under OPTIONS-AFTER and LETTERS-AFTER, and the candidate words of the neutral sentence under SENTENCE-AFTER.

Looked Up or Copied? Later Mentions Decide Which Attention Channel Carries an In-Context Value

Table 10. Key/value decomposition. Mean effects (nats of m_{SB}) of clamping the writing token’s key (d_K), value (d_V) or both (d_{KV}) to the source run’s from layer 0, measured against the self-clamp row; the interaction $d_{KV} - d_K - d_V$ with its 95% bootstrap CI; and the value identity ID_V (nats). $n = 150$ stories per format and model.

Model	Format	d_K	d_V	d_{KV}	interaction	ID_V
Qwen2.5-1.5B	LETTERS-AFTER	+9.9	+4.2	+14.5	+0.4 [-0.5, +1.3]	+0.7
	OPTIONS-AFTER	+6.3	+10.4	+16.7	-0.0 [-0.5, +0.5]	+5.0
	SENTENCE-AFTER	+0.5	+14.4	+14.8	-0.0 [-0.4, +0.3]	+6.5
	NO-MENTION	+1.8	+21.8	+23.5	-0.1 [-0.7, +0.5]	+10.2
	LIST-BEFORE	+1.2	+13.6	+14.5	-0.3 [-0.6, +0.1]	+5.9
Qwen2.5-3B	LETTERS-AFTER	+22.3	+0.8	+22.8	-0.4 [-1.1, +0.3]	+0.2
	OPTIONS-AFTER	+20.4	+9.0	+29.1	-0.4 [-1.1, +0.3]	+5.0
	SENTENCE-AFTER	+2.9	+26.8	+29.9	+0.2 [-0.6, +1.0]	+11.8
	NO-MENTION	+4.3	+41.1	+45.9	+0.5 [-0.1, +1.1]	+18.1
	LIST-BEFORE	+2.8	+23.4	+26.4	+0.2 [-0.5, +0.8]	+9.6
Mistral-7B	LETTERS-AFTER	+31.3	+1.5	+32.2	-0.5 [-1.2, +0.1]	+1.0
	OPTIONS-AFTER	+28.0	+2.6	+30.8	+0.1 [-0.6, +0.8]	+2.3
	SENTENCE-AFTER	+12.9	+19.5	+31.8	-0.6 [-2.3, +1.1]	+11.5
	NO-MENTION	+2.7	+28.8	+31.5	-0.1 [-0.4, +0.3]	+13.6
	LIST-BEFORE	+1.2	+30.9	+32.1	+0.1 [-0.3, +0.5]	+14.6
OLMo-2-7B	LETTERS-AFTER	+15.6	+0.1	+15.7	+0.0 [-0.1, +0.1]	+0.1
	OPTIONS-AFTER	+15.5	+5.8	+21.0	-0.3 [-0.9, +0.2]	+3.7
	SENTENCE-AFTER	+3.8	+16.8	+20.7	+0.0 [-0.5, +0.6]	+8.6
	NO-MENTION	+2.7	+23.5	+26.1	-0.1 [-0.5, +0.3]	+11.0
	LIST-BEFORE	+0.2	+23.6	+23.9	+0.1 [-0.1, +0.4]	+11.2
Qwen2.5-7B	LETTERS-AFTER	+40.2	+1.1	+41.2	-0.1 [-0.7, +0.4]	+0.2
	OPTIONS-AFTER	+39.6	+8.6	+47.9	-0.4 [-1.1, +0.3]	+5.8
	SENTENCE-AFTER	+11.0	+23.1	+34.4	+0.4 [-0.4, +1.1]	+11.9
	NO-MENTION	+6.0	+36.8	+42.5	-0.3 [-0.7, +0.2]	+16.6
	LIST-BEFORE	+2.4	+34.8	+36.8	-0.4 [-0.8, -0.1]	+16.0
Qwen3-8B	LETTERS-AFTER	+66.8	+1.2	+67.9	-0.1 [-0.5, +0.3]	+1.0
	OPTIONS-AFTER	+67.1	+13.2	+80.5	+0.2 [-1.4, +1.7]	+8.2
	SENTENCE-AFTER	+14.3	+40.0	+53.3	-1.1 [-2.5, +0.3]	+21.8
	NO-MENTION	+5.7	+54.4	+59.4	-0.7 [-1.6, +0.2]	+25.3
	LIST-BEFORE	+2.9	+51.8	+54.4	-0.3 [-0.8, +0.2]	+24.0
Qwen2.5-14B	LETTERS-AFTER	+70.0	+0.2	+70.2	+0.0 [-0.2, +0.3]	+0.1
	OPTIONS-AFTER	+75.0	+7.8	+82.8	-0.1 [-0.7, +0.5]	+5.4
	SENTENCE-AFTER	+21.9	+33.8	+56.9	+1.1 [-1.4, +3.6]	+18.5
	NO-MENTION	+10.8	+46.5	+57.1	-0.2 [-0.9, +0.4]	+19.1
	LIST-BEFORE	+2.4	+65.9	+68.3	+0.0 [-0.5, +0.5]	+31.1
Mistral-24B	LETTERS-AFTER	+18.7	+0.4	+19.0	-0.1 [-0.2, -0.0]	+0.2
	OPTIONS-AFTER	+17.1	+4.8	+21.9	+0.1 [-0.2, +0.4]	+3.6
	SENTENCE-AFTER	+8.1	+10.4	+18.6	+0.1 [-0.7, +0.8]	+6.8
	NO-MENTION	+2.0	+20.8	+22.6	-0.2 [-0.4, +0.0]	+9.8
	LIST-BEFORE	-0.0	+19.4	+19.4	+0.1 [-0.1, +0.3]	+9.1
Qwen2.5-32B	LETTERS-AFTER	+69.5	+0.6	+70.3	+0.2 [-0.1, +0.5]	+0.3
	OPTIONS-AFTER	+61.4	+10.2	+71.6	-0.1 [-1.8, +1.7]	+7.1
	SENTENCE-AFTER	+18.7	+44.8	+63.1	-0.4 [-2.0, +1.3]	+23.4
	NO-MENTION	+9.8	+53.8	+63.4	-0.2 [-0.9, +0.5]	+23.4
	LIST-BEFORE	+2.8	+65.1	+68.1	+0.1 [-0.5, +0.7]	+30.6
Qwen2.5-72B	LETTERS-AFTER	+52.8	+0.2	+53.0	-0.0 [-0.2, +0.2]	+0.3
	OPTIONS-AFTER	+44.6	+14.3	+59.6	+0.7 [-0.3, +1.7]	+10.5
	SENTENCE-AFTER	+14.4	+30.9	+46.4	+1.1 [-0.3, +2.6]	+17.5
	NO-MENTION	+3.1	+47.7	+50.9	+0.1 [-0.4, +0.5]	+22.9
	LIST-BEFORE	-1.7	+51.2	+49.8	+0.3 [-0.2, +0.7]	+24.1

Under SENTENCE-AFTER the “other later tokens” also recover a small part, and 0.19–0.22 of the effect is recovered by no single group. An earlier exploratory run at Qwen2.5-1.5B gives the same picture: the option words recover 0.87 of the key

Table 11. Key share by clamp onset. $s_K = \bar{d}_K / (\bar{d}_K + \bar{d}_V)$ under OPTIONS-AFTER with both clamps applied from layer ℓ_0 on; the onset used in each model is given in brackets. $n = 150$ stories per model; point estimates (the $\ell_0 = 0$ intervals are in Table 8). Exploratory.

Model	layers L	s_K , onset $\ell_0=0$	s_K , $\ell_0 \approx 0.06L$	s_K , $\ell_0 \approx 0.3L$
Qwen2.5-1.5B	28	0.37 ($\ell_0=0$)	0.37 ($\ell_0=2$)	0.19 ($\ell_0=8$)
Qwen2.5-3B	36	0.69 ($\ell_0=0$)	0.69 ($\ell_0=2$)	0.56 ($\ell_0=11$)
Mistral-7B	32	0.92 ($\ell_0=0$)	0.91 ($\ell_0=2$)	0.53 ($\ell_0=10$)
OLMo-2-7B	32	0.73 ($\ell_0=0$)	0.72 ($\ell_0=2$)	0.03 ($\ell_0=10$)
Qwen2.5-7B	28	0.82 ($\ell_0=0$)	0.82 ($\ell_0=2$)	0.66 ($\ell_0=8$)
Qwen3-8B	36	0.84 ($\ell_0=0$)	0.84 ($\ell_0=2$)	0.73 ($\ell_0=11$)
Qwen2.5-14B	48	0.91 ($\ell_0=0$)	0.91 ($\ell_0=3$)	0.83 ($\ell_0=14$)
Mistral-24B	40	0.78 ($\ell_0=0$)	0.78 ($\ell_0=2$)	0.37 ($\ell_0=12$)
Qwen2.5-32B	64	0.86 ($\ell_0=0$)	0.86 ($\ell_0=4$)	0.82 ($\ell_0=19$)
Qwen2.5-72B	80	0.76 ($\ell_0=0$)	0.76 ($\ell_0=5$)	0.55 ($\ell_0=24$)

Table 12. Depth and KV-group localisation of the option-word key read (Qwen2.5-1.5B, FP32). Entries are fractions of the effect obtained when the option-word rows see the source key in every layer and both KV groups, with 95% bootstrap CIs over stories (ratio of means recomputed in each resample). $n = 40$ stories per format. Exploratory.

Rows/layers that see the swapped key	OPTIONS-AFTER	LETTERS-AFTER
layers 0–3	-0.02 [-0.02, -0.01]	-0.01 [-0.01, -0.00]
layers 4–7	+0.23 [+0.11, +0.35]	+0.36 [+0.26, +0.46]
layers 8–11	+0.14 [+0.10, +0.18]	+0.06 [+0.03, +0.09]
layers 12–15	+0.15 [+0.12, +0.19]	+0.04 [+0.02, +0.06]
layers 16–19	+0.02 [+0.01, +0.04]	+0.01 [+0.00, +0.02]
layers 20–23	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
layers 24–27	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
KV group 0 (all layers)	+0.22 [+0.17, +0.27]	+0.06 [+0.03, +0.09]
KV group 1 (all layers)	+0.53 [+0.40, +0.66]	+0.66 [+0.57, +0.75]

effect under OPTIONS-AFTER and 0.98 under LETTERS-AFTER. Restricting the splice further, the read is concentrated in layers 4–15 of 28, and one of the two KV groups carries most of it (exploratory; Table 12).

Intervention frames. Table 16 gives every frame quantity with its interval. The removal fractions follow the addition fractions: ρ_K is large where ψ_K is, and ρ_V where ψ_V is. Among the six candidates, M’s top answer is the target T in 0.76–0.87 of (core, fit) items at Mistral-Small-24B and 0.92–0.95 at Qwen2.5-72B, depending on the format.

Competent-only analysis. Our primary analyses pool all stories, whether or not the model answers the clean runs correctly. As a secondary analysis, every natural-factorial quantity is also computed on the stories whose base and source clean runs are both answered correctly (argmax over the candidates). From 7B up most stories are competent and the two analyses agree closely. The competent subset is smallest at Qwen2.5-1.5B and 3B, and the estimates move most at 3B. The sign of ID_K under OPTIONS-AFTER, LETTERS-AFTER, NO-MENTION and LIST-BEFORE is unchanged in every model. The near-zero SENTENCE-AFTER value at Qwen2.5-3B becomes slightly positive, with a CI that excludes zero, but stays far below the SENTENCE-AFTER values from 7B up. No conclusion in the main text changes.

Attention probe. In an earlier configuration (Qwen2.5-1.5B, the question “Where is the $\{o\}$ actually located?”, options after the question, no system turn or prefill, $n = 25$ cores, no CIs), we measured head-averaged attention from each option word to the writing token, with the writing token’s key swapped to the source run’s in all layers. The option word that names the source location then attends to the writing token as it does in the source run, more than the non-matching option words do (results/choice_attention_qwen1.5b_world.txt). This probe is exploratory and uses a different prompt from the main experiments.

Table 13. **Paint and schedule tasks.** Key share under OPTIONS-AFTER and ID_K (nats) in each format, clamped from layer 0. $n = 150$ stories per task, format and model; 95% bootstrap CIs. Preregistration D (D1, D2) for OPTIONS-AFTER, NO-MENTION and LIST-BEFORE; the rest exploratory.

Task	Model	s_K (OPTIONS-AFTER)	ID_K (nats)				
			LETTERS-AFTER	OPTIONS-AFTER	SENTENCE-AFTER	NO-MENTION	LIST-BEFORE
paint	Qwen2.5-7B	0.73 [0.72, 0.74]	+21.5 [+21.3, +21.7]	+21.4 [+21.2, +21.7]	+3.0 [+2.8, +3.1]	+1.2 [+1.1, +1.3]	+0.3 [+0.2, +0.4]
	Qwen2.5-14B	0.90 [0.90, 0.91]	+35.7 [+35.4, +36.0]	+40.3 [+39.9, +40.8]	+3.5 [+3.3, +3.7]	+0.8 [+0.7, +0.9]	+0.2 [+0.1, +0.2]
	Qwen3-8B	0.92 [0.92, 0.93]	+39.7 [+39.4, +40.1]	+42.7 [+42.3, +43.0]	+4.9 [+4.7, +5.2]	+1.1 [+1.0, +1.2]	+0.1 [+0.1, +0.2]
	Mistral-7B	0.86 [0.85, 0.87]	+16.0 [+15.7, +16.3]	+16.1 [+15.9, +16.4]	+2.2 [+1.9, +2.4]	+0.4 [+0.3, +0.5]	+0.1 [+0.1, +0.2]
	OLMo-2-7B	0.47 [0.46, 0.48]	+7.4 [+7.2, +7.5]	+6.5 [+6.3, +6.7]	+0.7 [+0.6, +0.8]	+0.1 [+0.1, +0.2]	-0.1 [-0.1, -0.1]
schedule	Qwen2.5-7B	0.58 [0.56, 0.59]	+19.6 [+19.3, +20.0]	+12.7 [+12.4, +13.0]	+4.1 [+3.8, +4.4]	+0.1 [+0.1, +0.2]	-0.1 [-0.2, -0.1]
	Qwen2.5-14B	0.87 [0.87, 0.88]	+35.6 [+35.3, +35.9]	+26.1 [+25.7, +26.5]	+10.9 [+10.3, +11.6]	+0.6 [+0.5, +0.6]	+0.2 [+0.1, +0.2]
	Qwen3-8B	0.92 [0.91, 0.92]	+39.6 [+39.2, +40.0]	+28.5 [+28.0, +29.0]	+7.3 [+6.9, +7.7]	+0.9 [+0.8, +1.0]	+0.2 [+0.1, +0.2]
	Mistral-7B	0.82 [0.81, 0.84]	+16.2 [+15.6, +16.7]	+13.2 [+12.9, +13.5]	+5.1 [+4.6, +5.6]	+0.1 [+0.1, +0.2]	-0.0 [-0.1, +0.0]
	OLMo-2-7B	0.35 [0.34, 0.37]	+7.4 [+7.2, +7.6]	+4.5 [+4.3, +4.7]	+2.3 [+2.1, +2.4]	+0.1 [+0.1, +0.1]	+0.0 [-0.0, +0.1]

F. Further discussion

Relation to lookback, address and payload accounts. In the lookback of [Prakash et al. \(2026, Sec. 2\)](#), an ordering-ID pointer at a later token is matched through the query-key circuit to an address at the recalled token, here the state token (our writing token), and the payload stored there is copied forward through the OV circuit; [Wu & Shomali \(2026\)](#) likewise find that an operation’s cache is read mainly as an address rather than a payload. In our terms their answer lookback is a *copy* read, because the match is on ordering IDs and the state’s identity travels in the value, and our NO-MENTION results fit this payload side: the identity is copied from the writing token’s value. We do not test the address side, since ID_K counts only value-specific content and an ordering-ID address shared by base and source runs would not register; for the same reason the role-swap control, in which the observer is the first character named in both runs, cannot detect an ordering ID of the observer at the writing token. Later mentions add a second route, an identity lookup in which a re-mentioned word finds the writing token through its key, as duplicate-token heads do ([Wang et al., 2022](#)). We did not identify the heads, or test whether this route runs alongside an address-based lookback, as it might in the preliminary second dataset of [Prakash et al. \(2026\)](#), whose questions name both candidate states.

Table 14. Identity, not belief role. Role effect: mean change in $m = \log p(\text{initial}) - \log p(\text{LOC})$ when the observer and non-observer are swapped with the same location word (nats). f_K, f_V : fraction of that change reproduced by clamping only the writing token’s key or value (all layers) to the role-swapped run’s, as a ratio of means. Direct question; $n = 150$ stories per format and model; 95% bootstrap CIs. Preregistration E (E2) for OPTIONS-AFTER and NO-MENTION in the four 7–14B models; LETTERS-AFTER and Qwen2.5-72B exploratory.

Model	Format	role effect (nats)	f_K	f_V
Qwen2.5-7B	OPTIONS-AFTER	+34.4	+0.006 [+0.004, +0.008]	+0.010 [+0.007, +0.013]
	NO-MENTION	+28.7	+0.014 [+0.011, +0.018]	+0.001 [+0.000, +0.003]
	LETTERS-AFTER	+29.8	+0.003 [-0.000, +0.005]	+0.018 [+0.013, +0.025]
Qwen2.5-14B	OPTIONS-AFTER	+79.8	+0.001 [+0.000, +0.001]	-0.000 [-0.001, +0.000]
	NO-MENTION	+51.4	+0.001 [+0.000, +0.002]	-0.000 [-0.001, +0.001]
	LETTERS-AFTER	+66.0	-0.000 [-0.001, +0.000]	-0.000 [-0.001, +0.000]
Mistral-7B	OPTIONS-AFTER	+28.4	+0.003 [+0.002, +0.004]	+0.004 [+0.001, +0.008]
	NO-MENTION	+26.6	+0.019 [+0.014, +0.025]	+0.001 [-0.000, +0.003]
	LETTERS-AFTER	+28.9	+0.000 [-0.001, +0.002]	+0.011 [+0.006, +0.017]
OLMo-2-7B	OPTIONS-AFTER	+16.9	+0.004 [+0.003, +0.006]	+0.002 [+0.000, +0.003]
	NO-MENTION	+17.7	+0.014 [+0.011, +0.017]	+0.002 [-0.000, +0.004]
	LETTERS-AFTER	+12.5	-0.000 [-0.001, +0.001]	+0.003 [+0.001, +0.004]
Qwen2.5-72B	OPTIONS-AFTER	+50.6	+0.001 [+0.000, +0.002]	+0.002 [+0.001, +0.003]
	NO-MENTION	+38.4	+0.007 [+0.006, +0.009]	-0.001 [-0.002, +0.001]
	LETTERS-AFTER	+46.8	-0.001 [-0.002, -0.000]	+0.003 [+0.002, +0.004]

Table 15. Which later tokens read the writing token’s key. Fraction of the full key effect (all tokens see the source key, in every layer) recovered when only one group of later tokens sees it; ratio of mean log-odds changes. “Other later tokens” covers the instruction, chat template, prefill and answer position. Qwen2.5-7B, Qwen2.5-14B and Mistral-7B: BF16, $n = 60$ stories per format; preregistration D (D3) for OPTIONS-AFTER and SENTENCE-AFTER, LETTERS-AFTER exploratory. Qwen2.5-1.5B: FP32, $n = 40$, exploratory. 95% bootstrap CIs. The groups need not add up to one.

Model	Format	later-mention words	question	story after writing token	other later tokens
Qwen2.5-1.5B	OPTIONS-AFTER	+0.87 [+0.80, +0.95]	-0.00 [-0.01, +0.01]	-0.01 [-0.01, -0.00]	+0.00 [-0.04, +0.03]
Qwen2.5-1.5B	LETTERS-AFTER	+0.98 [+0.96, +1.00]	+0.01 [-0.00, +0.01]	-0.00 [-0.01, +0.00]	-0.00 [-0.00, +0.00]
Qwen2.5-7B	OPTIONS-AFTER	+0.92 [+0.90, +0.94]	-0.00 [-0.00, +0.00]	+0.00 [-0.00, +0.00]	+0.01 [+0.01, +0.02]
Qwen2.5-7B	LETTERS-AFTER	+1.00 [+0.99, +1.01]	+0.00 [-0.00, +0.00]	+0.00 [-0.00, +0.00]	+0.00 [+0.00, +0.00]
Qwen2.5-7B	SENTENCE-AFTER	+0.67 [+0.60, +0.73]	+0.00 [-0.00, +0.01]	-0.01 [-0.01, -0.00]	+0.11 [+0.09, +0.13]
Qwen2.5-14B	OPTIONS-AFTER	+0.94 [+0.93, +0.96]	-0.00 [-0.00, +0.00]	-0.00 [-0.00, +0.00]	+0.01 [+0.01, +0.01]
Qwen2.5-14B	LETTERS-AFTER	+1.00 [+1.00, +1.00]	-0.00 [-0.00, -0.00]	-0.00 [-0.00, +0.00]	-0.00 [-0.00, -0.00]
Qwen2.5-14B	SENTENCE-AFTER	+0.72 [+0.64, +0.80]	+0.00 [-0.00, +0.00]	-0.00 [-0.00, +0.00]	+0.09 [+0.07, +0.10]
Mistral-7B	OPTIONS-AFTER	+0.98 [+0.96, +0.99]	-0.00 [-0.00, +0.00]	-0.00 [-0.00, -0.00]	+0.01 [+0.00, +0.01]
Mistral-7B	LETTERS-AFTER	+1.00 [+1.00, +1.01]	+0.00 [+0.00, +0.00]	+0.00 [-0.00, +0.00]	+0.00 [-0.00, +0.00]
Mistral-7B	SENTENCE-AFTER	+0.73 [+0.67, +0.78]	+0.00 [-0.00, +0.00]	-0.00 [-0.00, +0.00]	+0.05 [+0.03, +0.06]

Table 16. The intervention of Anonymous (2026) across formats. φ : behavioural shift of the learned remap M relative to the PCA patch P, as a fraction of the natural source-to-target shift (Equation (4)). ψ_K , ψ_V : fraction of the M–P difference that P reproduces when given M’s keys or values at the writing token (Equations (5) and (6)). ρ_K , ρ_V : fraction of that difference that M loses when given P’s keys or values (Appendix C). Exchanges from block 6 on. $n = 96$ story cores with B, S and T distinct, three fits averaged within each core; 95% bootstrap CIs over cores. Preregistrations C (C1–C3) and E (E4) for the cells they name; the rest exploratory.

Model	Format	φ	ψ_K	ρ_K	ψ_V	ρ_V
Mistral-Small-24B	LETTERS-AFTER	0.75 [0.71, 0.79]	0.78 [0.73, 0.82]	0.99 [0.98, 0.99]	0.02 [0.01, 0.02]	0.16 [0.13, 0.19]
	OPTIONS-AFTER	0.70 [0.68, 0.73]	0.53 [0.49, 0.57]	0.76 [0.75, 0.78]	0.23 [0.22, 0.24]	0.44 [0.41, 0.48]
	SENTENCE-AFTER	0.72 [0.69, 0.75]	0.20 [0.16, 0.23]	0.45 [0.43, 0.47]	0.52 [0.50, 0.54]	0.77 [0.73, 0.81]
	NO-MENTION	0.71 [0.68, 0.73]	0.07 [0.06, 0.08]	0.18 [0.16, 0.20]	0.80 [0.78, 0.83]	0.93 [0.92, 0.94]
	LIST-BEFORE	0.74 [0.71, 0.76]	-0.00 [-0.01, 0.01]	0.01 [-0.00, 0.02]	0.97 [0.96, 0.98]	0.96 [0.95, 0.98]
Qwen2.5-72B	LETTERS-AFTER	0.90 [0.87, 0.93]	0.89 [0.86, 0.91]	0.99 [0.99, 1.00]	0.01 [0.00, 0.01]	0.17 [0.13, 0.20]
	OPTIONS-AFTER	0.89 [0.86, 0.92]	0.63 [0.59, 0.67]	0.79 [0.78, 0.81]	0.21 [0.19, 0.23]	0.44 [0.39, 0.48]
	SENTENCE-AFTER	0.87 [0.84, 0.89]	0.21 [0.17, 0.25]	0.34 [0.31, 0.36]	0.67 [0.65, 0.70]	0.80 [0.76, 0.84]
	NO-MENTION	0.86 [0.84, 0.88]	0.05 [0.05, 0.06]	0.11 [0.09, 0.12]	0.90 [0.88, 0.91]	0.95 [0.94, 0.96]
	LIST-BEFORE	0.91 [0.88, 0.92]	-0.03 [-0.04, -0.02]	-0.02 [-0.04, -0.01]	1.00 [0.98, 1.01]	1.01 [0.99, 1.02]